

Dual Recognition of Multiple Signals in Bacterial Outer Membrane Proteins Enhance Assembly and Maintain Membrane Integrity

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Abstract

Outer membrane proteins (OMPs) are essential components of the outer membrane of Gram-negative bacteria. In terms of protein targeting and assembly, the current dogma holds that a “ β -signal” imprinted in the final β -strand of the OMP engages the β -barrel assembly machinery (BAM complex) to initiate membrane insertion and assembly of the OMP into the outer membrane. Here, we reveal an additional rule, that signals equivalent to the β -signal are repeated in other, internal β -strands within bacterial OMPs. The internal signal is needed to promote the efficiency of the assembly reaction of these OMPs. BamD, an essential subunit of the BAM complex, recognizes the internal signal and the β -signal, arranging several β -strands for rapid OMP assembly. The internal signal-BamD ordering system is not essential for bacterial viability but is necessary to retain the integrity of the outer membrane against antibiotics and other environmental insults.

Teaser

Bacterial outer membrane proteins are recognized and bound by BamD at specific signals located in multiple β -strands at the C-terminus of these proteins.

eLife assessment

This **important** study reports the identification of a new amino acid sequence motif (i.e., "internal beta-signal") on outer membrane proteins, which is recognized by beta-assembly machinery in gram-negative bacteria. The authors carried out rigorous experiments, providing **compelling** evidence in support of their conclusions. This work significantly advances our understanding of the biogenesis of outer membrane proteins.

Introduction

In Gram-negative bacteria outer membrane proteins (OMPs) play important roles in membrane-mediated processes (Lundquist et al., 2021 [DOI](#); Tommassen, 2010 [DOI](#); Walther et al., 2009 [DOI](#)). Structurally, each major OMP has a β -barrel transmembrane domain, spanning the membrane as a cylindrical structure formed from amphiphilic, antiparallel β -strands (Konovalova et al., 2017 [DOI](#); Schulz, 2000 [DOI](#)). Starting as nascent polypeptides synthesized by ribosomes in the cytoplasm, these newly synthesized OMPs cross the inner membrane in a process mediated by the SecYEG translocon (Manting et al., 2000 [DOI](#); Mori and Ito, 2001 [DOI](#)), and then traverse the periplasm supported by the periplasmic chaperones (Wang et al., 2021 [DOI](#)). The molecular machine driving OMP insertion and assembly into the membrane is the β -barrel Assembly Machinery complex (BAM complex) (Bakelar et al., 2016 [DOI](#); Gu et al., 2016 [DOI](#); Iadanza et al., 2016 [DOI](#)). Given this essential role, and its location on the bacterial cell surface, the BAM complex has emerged as an attractive antibiotic target (Hart et al., 2019 [DOI](#); Imai et al., 2019 [DOI](#); Kaur et al., 2021 [DOI](#); Luther et al., 2019 [DOI](#)).

In *E. coli*, the BAM complex is composed of five subunits BamA, BamB, BamC, BamD and BamE (Bakelar et al., 2016 [DOI](#); Gu et al., 2016 [DOI](#); Iadanza et al., 2016 [DOI](#); Wu et al., 2005 [DOI](#)). BamA, the essential core-subunit, is embedded in the outer membrane and projects an extensive N-terminal domain consisting of five polypeptide transport-associated (POTRA) repeats into the periplasm which interacts with other subunits (Noinaj et al., 2013 [DOI](#); Voulhoux, 2003 [DOI](#)). There is some variation in the lipoprotein subunits that are present in the BAM complex varies in different bacterial lineages (Anwari et al., 2012 [DOI](#); Webb et al., 2012 [DOI](#)) but both BamA and the lipoprotein BamD are found in the BAM complexes of all bacterial lineages and are essential for viability in *E. coli* (Malinverni et al., 2006 [DOI](#); Webb et al., 2012 [DOI](#); Wu et al., 2005 [DOI](#)). The interaction of the POTRA domains of BamA with its partner BamD contributes to a large, funnel-like feature of the BAM complex which expands into the periplasm.

BamA belongs to Omp85-superfamily protein conserved across bacterial lineages and organelles (Heinz and Lithgow, 2014 [DOI](#); Wu et al., 2005 [DOI](#)). Omp85-superfamily proteins are themselves β -barrels, utilizing a transiently open lateral gate between the first and last β -strands as the catalytic site of the OMPs assembly (Doyle et al., 2022 [DOI](#); Doyle and Bernstein, 2021 [DOI](#), 2019 [DOI](#); Noinaj et al., 2013 [DOI](#); Takeda et al., 2021 [DOI](#); Tomasek et al., 2020 [DOI](#); Xiao et al., 2021 [DOI](#)). Biophysical studies have shown "snap-shot" structures of assembly intermediates of the BAM complex with substrate OMPs and demonstrated that during assembly the first strand of the lateral gate of the BamA interacts with the C-terminal strand of the substrate OMPs (Doyle et al., 2022 [DOI](#); Shen et al., 2023 [DOI](#); Takeda et al., 2023 [DOI](#);

Tomasek et al., 2020 [DOI](#); Wu et al., 2021 [DOI](#)). OMPs contain a conserved sequence termed the " β -signal" located within this C-terminus strand (Kutik et al., 2008 [DOI](#); Paramasivam et al., 2012 [DOI](#); Struyvé et al., 1991 [DOI](#); Wang et al., 2021 [DOI](#)). Inhibitory compounds such as darobactin show a

similar structure to that of the β -signal and are capable of interacting with the lateral gate (Imai et al., 2019; Kaur et al., 2021; Xiao et al., 2021). The current dogma suggested by these studies dictate that this β -signal should engage directly with the lateral gate in BamA (Horne and Radford, 2022; Tomasek and Kahne, 2021).

Despite recent advances in resolving the BAM complex structure and its function, the molecular mechanism of early stage OMP assembly remains enigmatic. It has been shown that the β -signal plays a significant role in the targeting and folding of BAM complex substrates (Kutik et al., 2008; Wang et al., 2021), yet several important OMPs do not have a recognizable β -signal motif in their C-terminal sequence (Celik et al., 2012; Hagan et al., 2015). For instance, BamA itself lacks an apparent C-terminal β -signal (Hagan et al., 2015). Also, a comprehensive study of the autotransporter class of OMPs, including proteins such as EspP, showed that while 1038 non-redundant proteins did have a C-terminal β -signal motif, over 40% of these proteins had a β -signal motif only in internal strands (Celik et al., 2012), which was also suggested to be the case for BamA (Hagan et al., 2015). It remains to be tested if these internal motifs function as β -signals and how they engage the BAM complex.

Before substrate OMPs can enter the lateral gate of BamA, they must first pass through the periplasmic features of the BAM complex. Although the assembly reactions that occur at the lateral gate are well understood, less is known about the molecular mechanisms at the early stage, prior to being position in the lateral gate. The specific points that have been established are that the substrate makes contact with the BamB and BamD subunits and the lateral gate of the BamA subunit (Hagan et al., 2013; Harrison, 1996; Ieva et al., 2011).

Here, we establish a screening system based on an *in vitro* reconstituted membrane assay using a translocation-competent *E. coli* Microsomal Membrane (EMM) (Gunasinghe et al., 2018; Thewasano et al., 2023) and demonstrate that OMPs - including the major porins that are the predominant substrate of the BAM complex - contain multiple β -signal related repeats. Comparison of the C-terminal β -signal with what we refer to here as the “internal signal”, revealed the essential elements of sequence and orientation in the signals that are recognized by the BAM complex. Internal signals and β -signals are recognized by BamD and are responsible for rapid assembly of the OMP into the bacterial membrane at the early stage. Moreover, *in situ* crosslinking and mutagenesis demonstrated that the specific relationship between the BAM complex and this dual signal plays an important role in the integrity of the outer membrane.

Results

Peptidomimetics derived from *E. coli* OmpC inhibit OMP assembly

The EMM assembly assay provides a means for *in vitro* reconstitution of BAM complex function, where the assembly of a ^{35}S -labelled OMP can be monitored (fig. S1A) (Gunasinghe et al., 2018; Thewasano et al., 2023). EspP, a model OMP substrate, belongs to autotransporter family of proteins. Autotransporters have two domains; (1) a β -barrel domain, assembled into the outer membrane via the BAM complex, and (2) a passenger domain, which traverses the outer membrane via the lumen of the β -barrel domain itself and is subsequently cleaved by the correctly assembled β -barrel domain (Celik et al., 2012). When EspP is correctly assembled into outer membrane, a visible decrease in the molecular mass of the protein is observed due to the self-proteolysis. Once the barrel domain is assembled into the membrane it becomes protease-resistant, with residual unassembled and passenger domains degraded (Leyton et al., 2014; Roman-Hernandez et al., 2014). When applied to ^{35}S -labelled EspP, the assay discriminates assembly of a protease-protected β -barrel domain, assembled into the outer membrane by the BAM complex (Ieva et al., 2011), from the protease-sensitive extracellular passenger domain (Fig. 1A) that are degraded by protease (fig. S1B). The addition of peptides with BAM complex

affinity, such as the OMP β -signal, are capable of exerting an inhibitory effect by competing for binding of substrate OMPs to the BAM complex (Hagan et al., 2015 [↗](#)). Thus, the addition of peptides derived from the entirety of OMPs to the EMM assembly assay, which can evaluate assembly efficiency with high accuracy, expects to identify novel regions that have affinity for the BAM complex. To screen for potential peptidomimetics that could inhibit BAM complex function, a synthetic peptide library was designed for coverage of the sequence features in OmpC (fig. S2). We used peptides that mapped the entirety of OmpC, with a two amino acid overlap. This we considered preferable to peptides that were restricted by structural features, such as β -strands, in consideration that β -strand formation may or may not have occurred in early-stage interactions at the BAM complex.

Six peptides (4, 10, 17, 18, 21, and 23) were found to inhibit EspP assembly (**Fig. 1A** [↗](#)). Of these, peptide 23 corresponds to the canonical β -signal of OMPs: it is the final β -strand of OmpC and it contains the consensus motif of the β -signal (ζ xGxx[Ω/Φ]x[Ω/Φ]). The inhibition seen with peptide 23 indicated that our peptidomimetics screening system using EspP can detect signals recognized by the BAM complex. In addition to inhibiting EspP assembly, five of the most potent peptides (4, 17, 18, 21, and 23) inhibited additional model OMPs; the porins OmpC and OmpF, the peptidoglycan-binding OmpA, and the maltoporin LamB (fig. S3). Comparing the sequences of these inhibitory peptides suggested the presence of a sub-motif from within the β -signal, namely [Ω/Φ]x[Ω/Φ] (**Fig. 1B** [↗](#)). The sequence codes refer to conserved residues such that: ζ , is any polar residue; G is a glycine residue; Ω is any aromatic residue; Φ is any hydrophobic residue and x is any residue (Hagan et al., 2015 [↗](#); Kutik et al., 2008 [↗](#)). The non-inhibitory peptide 9 contained some elements of the β -signal but did not show inhibition of EspP assembly (**Fig. 1A** [↗](#)).

Peptide 18 also showed a strong sequence similarity to the consensus motif of the β -signal (**Fig. 1B** [↗](#)) and, like peptide 23, had a strong inhibitory action on EspP assembly (**Fig. 1A** [↗](#)). Variant peptides based on the peptide 18 sequence were constructed and tested in the EMM assembly assay (**Fig. 1C** [↗](#)). This analysis revealed that the position 0 (0Φ), in addition to the [Ω/Φ]x[Ω/Φ], motif was functionally important to inhibitory action (**Fig. 1C** [↗](#)). Thus, we hypothesized the elements of an internal β -signal (hereafter the internal signal) to be Φ xxxxx[Ω/Φ]x[Ω/Φ].

Mutations to a putative internal signal results in loss of OMP assembly

To assess the impact of the conserved residues in the potential internal signal for OMP assembly, a series of mutant OmpC variants were constructed and assayed for assembly in the EMM assay (**Fig. 2A** [↗](#)). Three of the peptide regions are present in the final 5 β -strands of OmpC. We denoted these as the first (-1), third (-3), and fifth (-5) β strands from the C-terminus of the barrel. Note, since peptide 17 and peptide 18 overlap, they were considered together as a single region. Given the antiparallel structure of a β -barrel the -1, -3 and -5 strands are orientated in the same direction, and these three strands all conform to the putative Φ xxxxx[Ω/Φ]x[Ω/Φ] motif.

BN-PAGE analysis was employed to measure the efficiency of OmpC trimer assembly (**Fig. 2A** [↗](#)). These gels separate membrane proteins that have been solubilized from native membranes, allowing time-course analysis of protein assembly. Among the mutants with significantly reduced assembly efficiency, Y286A, Y325A, and Y365A correspond to Ω/Φ in position 6 (Ω/Φ), and F280A corresponds to Φ in position 0 of the putative signal (**Fig. 2A and fig** [↗](#). S4A). In terms of the highly conserved residues present, sequence conservation was mapped against the β -strands of OMPs whose structures have been determined (fig. S5, **S6** [↗](#), and S7). Analysis showed the -5 strand of OmpF and LamB includes residues that conform to the putative Φ xxxxx[Ω/Φ]x[Ω/Φ] motif (**Fig. 2B** [↗](#)). We constructed mutants to the 0Φ or 6[Ω/Φ] positions of the -5 strand in OmpF and LamB, these are the OmpF(V279A), OmpF(Y285A), LamB(V333A) and LamB(Y339A) mutants (**Fig. 2B** [↗](#)). BN-PAGE analysis showed that both the Φ – OmpF(V279) and LamB(V333) - and Ω/Φ – OmpF(Y285) and LamB(Y339) - residues were important for the assembly of OmpF and LamB (fig. S4B, C).

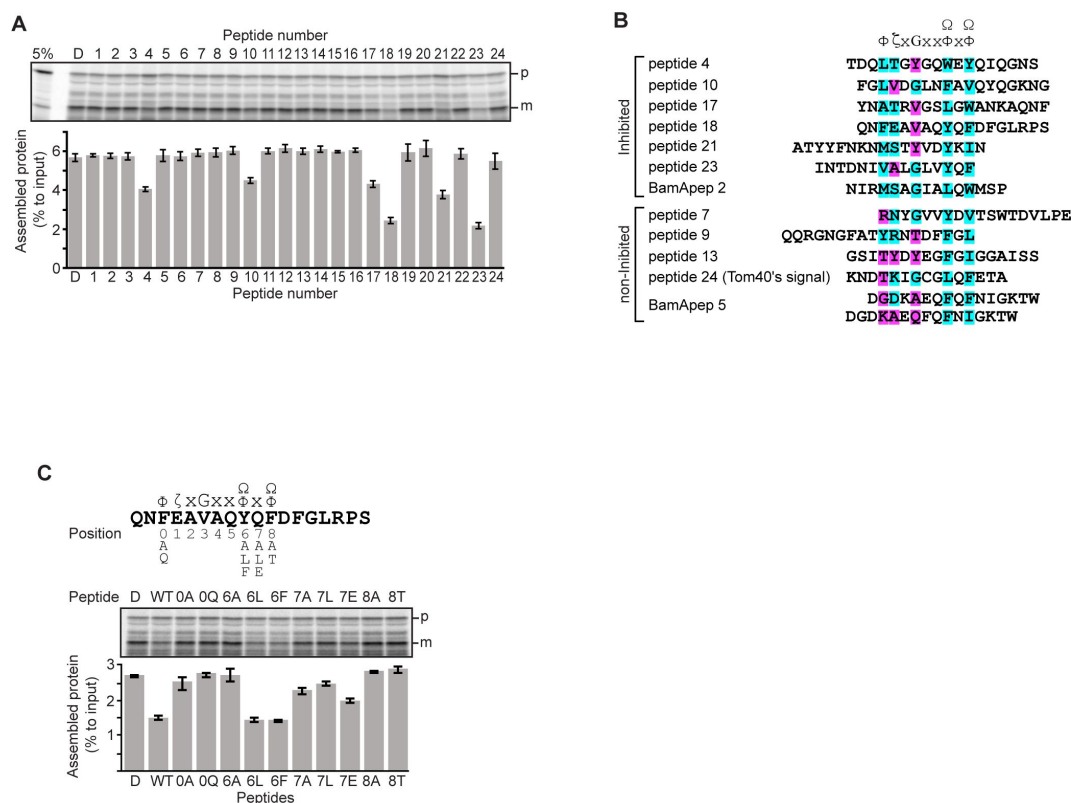


Fig. 1

Peptides derived from

β -strands of OmpC selectively inhibit OMP assembly. **A and C,** ^{35}S -labelled EspP was synthesized by rabbit reticulocyte lysate, then incubated with EMM, and samples prepared for analysis by SDS-PAGE and radio-imaging (see Methods). EspP assembly into EMM was performed in the presence of DMSO (D) or specific inhibitor peptides. The precursor (p) and mature (m) form of EspP are indicated. **A,** Peptides labelled 1 through 23 were derived from OmpC, while peptide 24 was taken from a mitochondrial β -barrel protein to serve as a negative control (see fig. S2). **B,** Sequence comparison of inhibitor peptides identified in the EspP assembly assays. The signal is comprised of key hydrophobic (Φ), polar (ζ), glycine (G), and 2 aromatic (Ω) or hydrophobic amino acids. Amino acids highlighted in blue conform to the consensus signal, while those highlighted in purple deviate from the consensus signal. **C,** Upper: sequence of the peptide 18 indicating residue codes for the mutated peptides e.g. at position "6", the native residue Y was mutated to A in peptide "6A". Position 0 designates the first hydrophobic residue and position 8 designates the final aromatic or hydrophobic residue. DMSO (D) was used for control. Middle: SDS-PAGE of EspP assembly assay. Lower: data for inhibition of EspP assembly into EMM assayed in the presence of each of the indicated peptides.

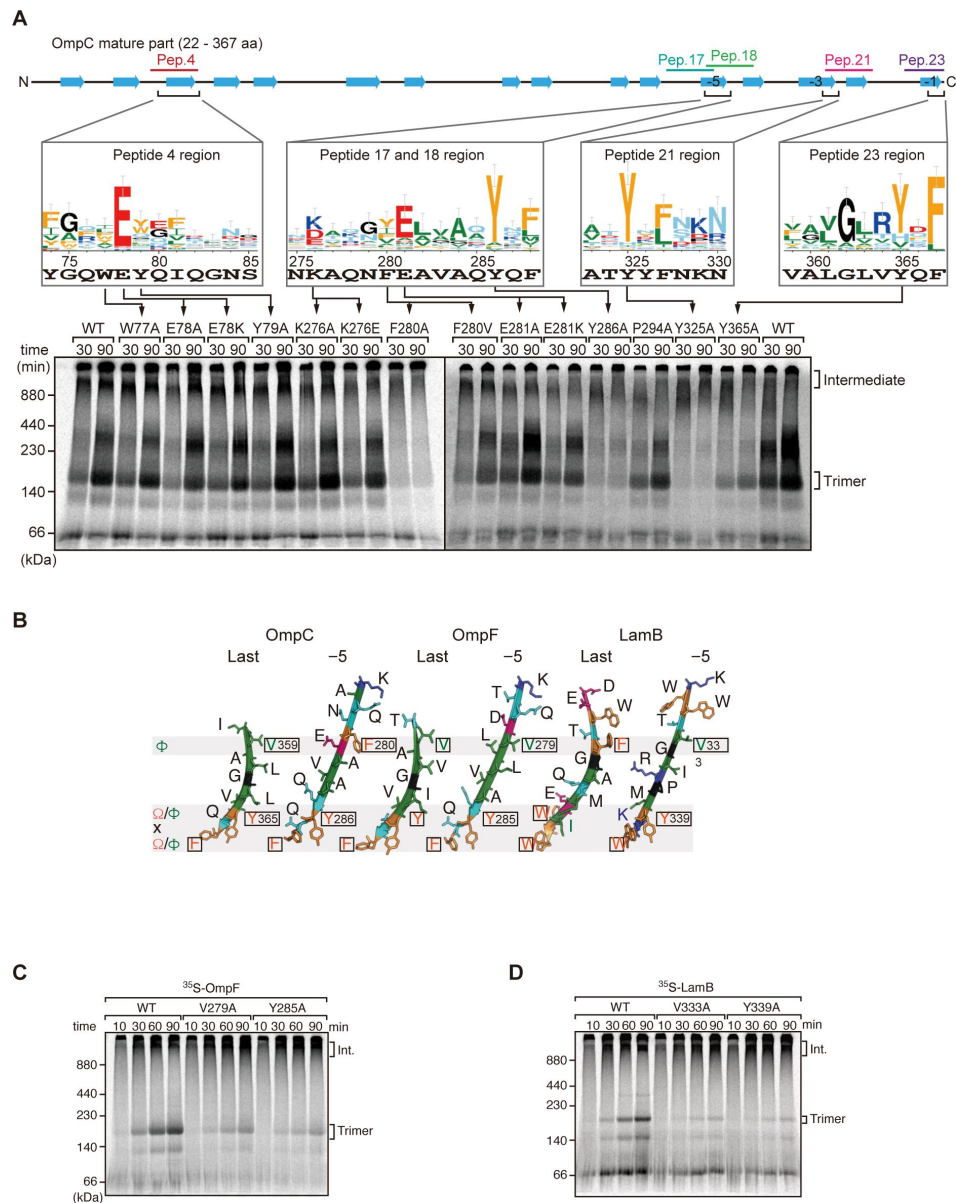


Fig. 2

Mutations in β -signals inhibit OMP Assembly *in vitro*. A, Representation of OmpC, (upper panel) annotated to show the position of each β -strand (blue arrows) and the region mimicked in the indicated peptides. Sequence logos show the conservation of sequence for the residues in these regions (see Methods), where the height of letter corresponds to the degree of conservation across bacterial species, residues are color-coded: aromatic (orange), hydrophobic (green), basic (blue), acidic (red), polar (sky blue) and proline and glycine (black). In the lower panel, data from the EMM assembly assay of OmpC mutants is organized to show effects for key mutants: ³⁵S-labelled OmpC wild-type or mutant proteins were incubated with EMM for 30 or 90 min, and analyzed by BN-PAGE and radio-imaging. B, Comparison of amino acids in the final and 5th to final (-5) β -strands of OmpC, OmpF, and LamB. All three OMPs contain the putative hydrophobic residue near the N-terminal region of the β -strands, highlight with a black box. A hydrophobic and an aromatic residue is also found to be highly conserved between these OMP β -strands at the C-terminal amino acids of the β -strands. C-D, Assembly of ³⁵S-labelled OmpF (C) or LamB (D) with mutations to the -5 β -strand or final β -strand in to EMM. Samples were prepared as in A.

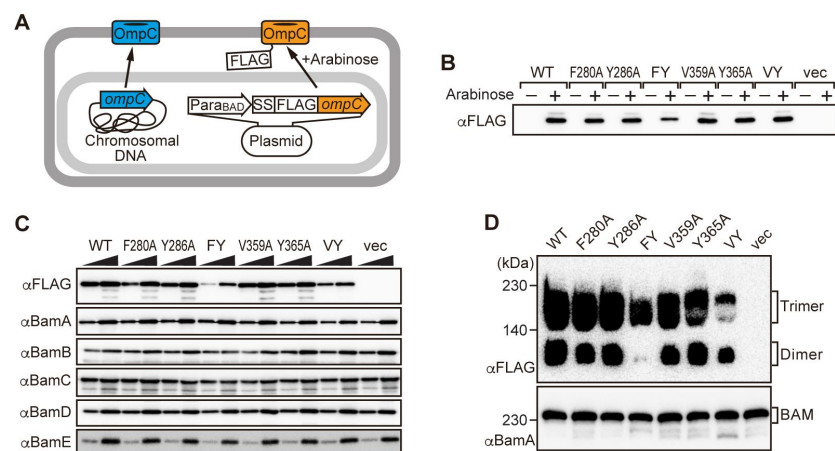


Fig. 3

Modifications in the putative internal signal slow the rate of OMP assembly *in vivo*. **A**, Schematic model of FLAG-OmpC expression system in *E. coli*. The plasmid-borne copy of OmpC was mutated and the mutant proteins are selectively detected with antibodies to the FLAG epitope. **B**, Total cell lysates were prepared from the variant *E. coli* strains expressing FLAG-OmpC (WT), -5 strand mutants: F280A, Y286A, double mutant F280A/Y286A (FY), or final β -strand mutants: V359A, Y365A, double mutant V359A/Y365A (VY) with (+) or without (-) arabinose. The steady-state levels of the indicated proteins were assessed after SDS-PAGE by immunoblotting with anti-FLAG antibody. **C**, After growth with arabinose induction, EMM fraction was isolated from each of the FLAG-OmpC variant expressing *E. coli* strains to assess steady state levels of OmpC variants and BAM complex subunits: BamA, BamB, BamC, BamD, and BamE by SDS-PAGE and immunoblotting. **D**, EMM fractions were solubilized with 1.5% DDM and subjected to BN-PAGE and immunoblotting. The indicated proteins were detected by anti-FLAG (top) and anti-BamA antibodies (bottom), respectively.

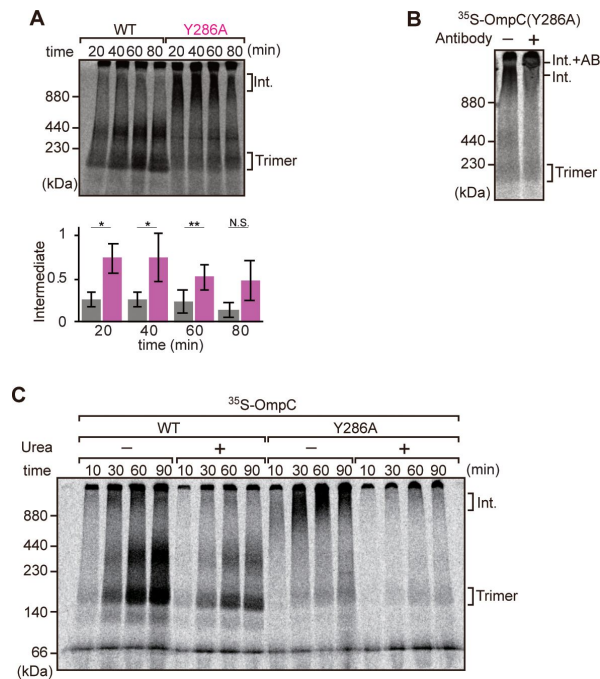


Fig. 4

Mutation that stalls OMP assembly in a high molecular weight assembly intermediate. **A**, Upper panel, assembly assay of ³⁵S-OmpC wild-type (WT) and mutant Y286A. After incubation in the EMM assays for indicated time points, ³⁵S-proteins were resolved via BN-PAGE. OmpC trimer and intermediate (int) are indicated. The lower panel shows a quantification of the intermediate form compared to total assembled protein (average of 3 independent experiments). Statistical significance was indicated by the following: N.S. (not significant), $p < 0.05$; *, $p < 0.005$; **. Exact p values of intermediate formed by Wt vs Y286A at each timepoint were as follows; 20 minutes: $p = 0.03077$, 40 minutes: $p = 0.02402$, 60 minutes: $p = 0.00181$, 80 minutes: $p = 0.0545$. **B**, ³⁵S-labelled OmpC(Y286A) was incubated with EMM for 60 mins, then the EMM fraction was solubilized with 1.5% DDM and incubated with (+) or without (-) anti-BamC antibody for 40 min. These samples were then analyzed by BN-PAGE. The presumptive translocation intermediate (Int) and the gel-shift species (Int. +AB) are indicated. **C**, ³⁵S-labelled OmpC was incubated with EMM and then analyzed by BN-PAGE and radio-imaging. EMM were then washed and solubilized in the presence (+) or absence (-) of 6 M urea, after which the membrane fraction was re-isolated via ultracentrifugation for 45 min. Isolated membranes were solubilized in 1.5% DDM and analyzed by BN-PAGE.

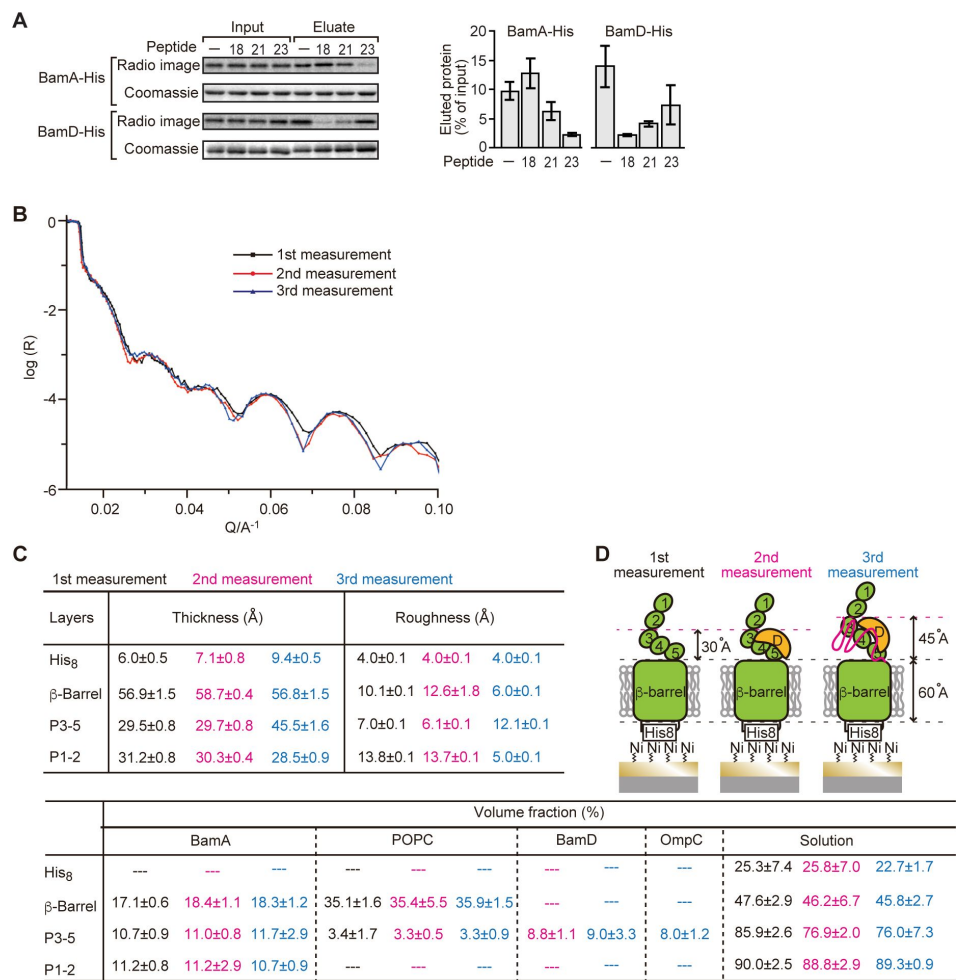


Fig. 5

Proximity and dependence of BamD with OmpC(Y286A) revealed by Neutron Reflectometry analysis. **A**, Ni-NTA was used to affinity purify His6-BamA or His6-BamD. The protein-Ni-NTA bead complex were then incubated in the presence of DMSO or the indicated peptides resuspended in DMSO, and ³⁵S-labelled OmpC was incubated with these beads. After washing, bound proteins were eluted with 400 mM imidazole buffer. The graph represents densitometry analysis of data from 3 independent experiments. **B**, NR profiles for BamA in membrane (1st measurement, square black line), after addition of BamD (2nd measurement, circle red line), and after addition of the OmpC(Y286A) substrate (3rd measurement, triangle blue line). The experimental data was fitted using a seven-layer model: chromium - gold - NTA - His₈ - β-barrel - P3-5 - P1-2. **C**, Table summarizes of the thickness, roughness and volume fraction data of each layer from the NR analysis. The thickness refers to the depth of layered structures being studied as measured in Å. The roughness refers to the irregularities in the surface of the layered structures being studied as measured in Å. 1st, 2nd, and 3rd measurement displayed in black, magenta, and blue, respectively. P3-5: POTRA3, POTRA 4, and POTRA5, P1-2: POTRA1 and POTRA2, Lipid: POPC, Solution: D₂O, gold match water (GMW) and H₂O. Details were described in Table S8 to S10. **D**, Neutron reflectometry schematic of BamA (green), BamD (yellow) and OmpC(Y286A) (magenta). Numbers indicate corresponding POTRA domains of BamA. Note that the cartoon is a depiction of the results and not a scale drawing of the structures.

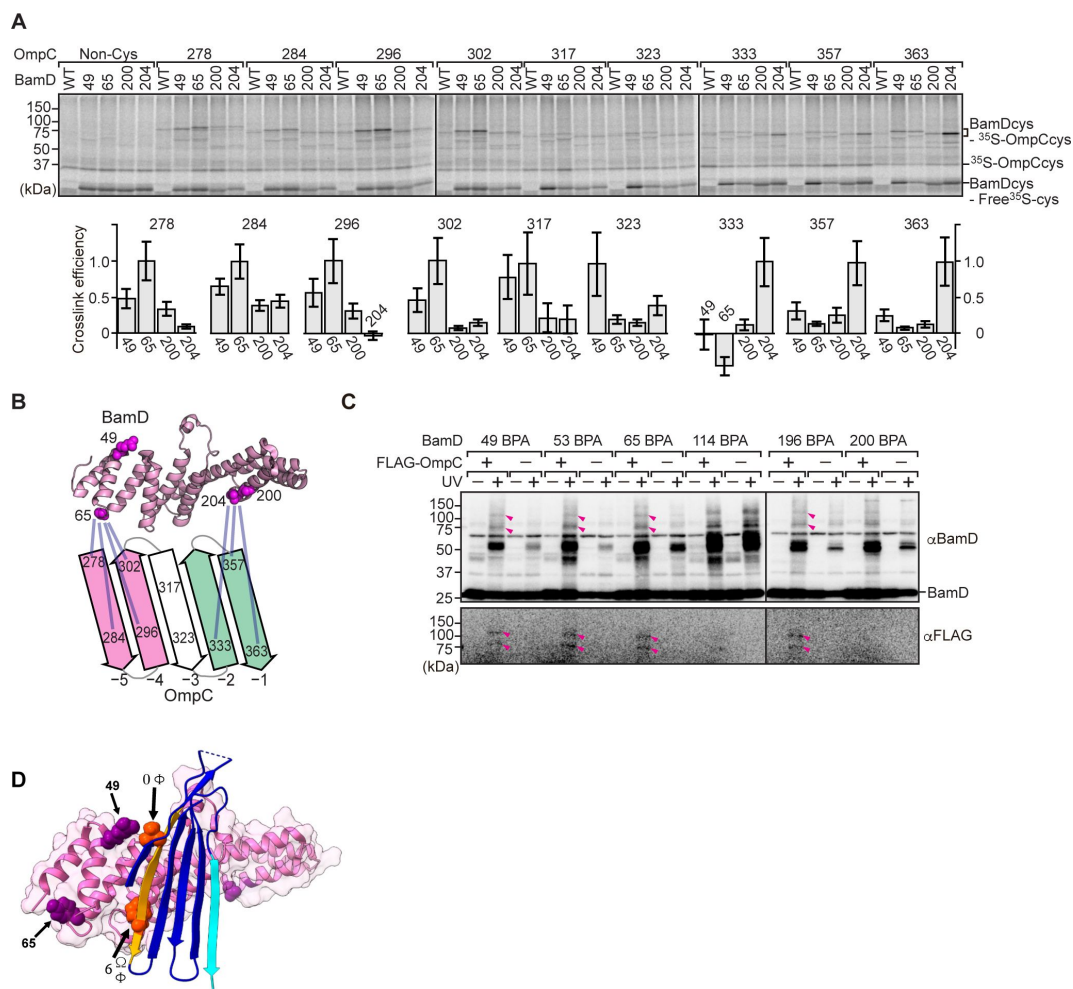


Fig. 6

BamD acts as the receptor for internal β -signals. **A**, BamD-cys mutant proteins were purified and incubated with ^{35}S -OmpC with cysteine mutations from the -5 to the final β -strand. After the addition of the oxidizing agent, CuSO_4 , samples were purified with Ni-NTA and analyzed by SDS-PAGE. **B**, BamD cysteine mutants, highlighted in magenta, formed disulfide bonds at distinct locations of OmpC. **C**, BPA containing BamD was expressed in the presence (+) or absence (-) of FLAG-OmpC overexpression. Pink arrows indicate BamD-FLAG-OmpC cross-link products. **D**, OMP substrate superimposed on BamD. BamD cross-linking positions corresponding to internal signal are marked in purple and substrate cross-linking regions are marked in orange.

To address whether the results seen *in vitro* in the EMM assays were consistent with phenotypes drawn from intact *E. coli* cells, we constructed arabinose-inducible plasmids to express mutants of Φ (position 0), the $[\Omega/\Phi]$ (position 6) and a double mutant combination of both positions in either the -5 or -1 strands of OmpC. To distinguish these plasmid-encoded OmpC variants from the endogenous, chromosome-encoded OmpC, a FLAG epitope tag was incorporated into the region of OmpC immediately behind the cleavage site for the secretory signal sequence (**Fig. 3A**) (Rapoport, 2007). Expression of FLAG-OmpC variants was induced with 0.1% arabinose, as confirmed through immunoblotting of total cell lysate (**Fig. 3B**). In comparing the *E. coli* strains, a marked reduction was seen in the steady-state level of the double (“FY” or “VY”) mutants OmpC (F280A+Y286A or V359A+Y365A) (**Fig. 3C**). Little change was evident in the expression levels of the single -1 mutants, suggesting that a fully conserved β -signal located at the C-terminus alone is not essential to maintaining OmpC assembly. To further investigate the nature of the defects, the various strains were analyzed BN-PAGE. In all cases, OmpC that has been detergent-solubilized from outer membranes migrates as a characteristic population of dimeric and trimeric forms under conditions of the BN-PAGE (**Fig. 3D**). Both the trimeric and dimeric forms of OmpC solubilized from the β -barrel arrays in the outer membrane have equivalent structural and functional properties (Rocque and McGroarty, 1989). The double (“FY” or “VY”) mutants displayed drastic reductions of the dimer and trimer (**Fig. 3D**). As a control, the level and integrity of the BAM complex were monitored and found to be equivalent in all strains (**Fig. 3C**, and D). Taken together these results suggest either the internal signal (in strand -5) or the C-terminal β -signal (in strand -1) must be present to achieve efficient assembly of OmpC into the outer membrane.

The internal signal is necessary for insertion step of assembly into OM

The β -signal located in the C-terminal β -strand is important for recognition by the BAM complex. In addition, because it is the last β -strand, it also contributes to engagement with the first β -strand in order that purified OMPs can fold into β -barrels, even in the absence of the BAM complex in artificial non-membrane environments such as detergent micelles (Burgess et al., 2008). To compare this property of strand-engagement for β -barrel folding in detergent micelles, we purified urea-denatured OmpC and the OmpC variant proteins with mutations in the -5 strand (F280A and Y286A) and the -1 strand (V359A and Y365A). The spontaneous folding of each OmpC variant was assessed by the rate of formation of trimer in the presence of detergent micelles (fig.S8A). Trimer embedded into detergent micelles migrated at 100 kDa in this assay system, whereas non-folded monomeric OmpC migrated at 37 kDa (fig. S8). Relative to the rate/extent of trimer formation observed for OmpC(WT), mutations to the $[\Omega/\Phi]$ (position 6) in the -5 strand i.e. OmpC(Y286A) or the -1 strand i.e. OmpC(Y365A) showed significantly impaired spontaneous folding. Position 6 is an amino acid that contributes to the aromatic girdle. Since Y286A and Y365A affected OMP folding as measured in folding experiments, their positioning into the aromatic girdle may contribute to the efficiency of β -barrel folding, in addition to contributing to the internal signal. Mutation of the Φ residue (position 0) in the -5 strand i.e. OmpC(F280A) or the -1 strand i.e.

OmpC(V359A) maintained a similar ability to assemble into micelles as native OmpC. Thus, the conserved features of both -1 and -5 strands have equivalent impact on intrinsic properties for spontaneous folding of β -barrel of OmpC.

We further analyzed the role of internal β -signal by the EMM assembly assay.

Compared to the rate at which the trimeric form of OmpC is assembled (**Fig. 4A**), OmpC(Y286A) is slow and the gels resolve that most of the ^{35}S -labelled OmpC(Y286A) is held in a high-molecular-weight species for most of the time-course. Only at 80 min of assembly has substantial amount of the ^{35}S -labelled OmpC(Y286A) been resolved from this assembly intermediate (“Int”) to a native,

trimeric form (**Fig. 4A** [↗](#)). In analyzing assembly intermediates of other sorts of membrane proteins in mitochondria, a version of this analysis referred to as “gel shift” BN-PAGE analysis has been developed ([Shiota et al., 2012](#) [↗](#)). Here, the addition of an antibody recognizing the surface-exposed BamC to the samples prior to electrophoresis resulted in a dramatic shift *i.e.* retardation of electrophoretic mobility due to the added mass of antibodies. This is consistent with OmpC(Y286A) being held in an assembly intermediate engaged with the BAM complex (**Fig. 4B** [↗](#)). Consistent with this, the assembly intermediate which was prominently observed at the OmpC(Y286A) can be extracted from the membranes with urea; urea extraction of the membrane samples prior to analysis by BN-PAGE showed that the integral membrane trimer form of OmpC and OmpC(Y286A) were not extracted by urea (**Fig. 4C** [↗](#)). Conversely, the assembly intermediate was completely extracted with urea, as expected for a substrate engaged in protein-protein interactions with a proteinaceous complex in the membrane fraction. These results suggest that the mutation in OmpC(Y286A) causes a longer interaction with the BAM complex, as a species of protein that is not yet inserted into the surrounding membrane environment.

Three subunits of the BAM complex have been previously shown to interact with the substrates: BamA, BamB, and BamD ([Hagan et al., 2013](#) [↗](#); [Harrison, 1996](#) [↗](#); [Ieva et al., 2011](#) [↗](#)). *In vitro* pull-down assay showed that while BamA and BamD can independently bind to the *in vitro* translated OmpC polypeptide (**Fig. S9A**), BamB did not (**Fig. S9B**). To test for signal-dependent binding, BamA and BamD were pre-incubated with inhibitor peptides (**Fig. 5A** [↗](#)). Peptide 23 inhibited binding of the substrate by BamA and peptide 18 inhibited binding of the substrate by BamD. Peptide 21 inhibited binding of the substrate by either BamA or BamD. These results suggest that BamA and BamD are important for recognition of both the β -signal and the internal signal. While the biochemical assay demonstrated that the OmpC(Y286A) mutant forms a stalled intermediate with the BAM complex, in a state in which membrane insertion was not completed, biochemical assays such as this cannot elucidate where on BamA-BamD this OmpC(Y286A) substrate is stalled. Neutron reflectometry (NR) is a powerful tool for probing the details of substrate- chaperone interactions, and is ideally suited for membrane-embedded proteins ([Shen et al., 2014](#) [↗](#)). The rationale to these experiments is that NR provides: (i) information on the distance of specified subunits of a protein complex away from the atomically flat gold surface to which the complex is attached, and (ii) allows the addition of samples between measurements, so that multi-step changes can be made to, for example, detect changes in domain conformation in response to the addition of a substrate. To apply NR imaging to monitor the engagement of OmpC with the BAM complex, BamA was immobilized on an Ni-NTA atomic flat gold-coated silicon wafer to orient the POTRA domain distal to the silicon wafer as previously described ([Ding et al., 2020](#) [↗](#)). A lipid layer was reconstituted to provide BamA with a membrane-like environment prior to sequential measurement of: (i) BamA with lipid (1st measurement), (ii) BamA with lipid and BamD (2nd measurement), and (iii) BamA-BamD-lipid with the addition of OmpC (WT) or OmpC(Y286A) (3rd measurement) (**fig. S10A**).

Comparing the reflectivity profiles of three measurements (**Fig. 5B** [↗](#)), changes were observed in the layers after the addition of BamD and OmpC(Y286A), as seen in a shift of fringe to low Q range around $0.02\text{--}0.03\text{ \AA}^{-1}$. Crystal structures show that the periplasmic domains of BamA can be treated as two rigid bodies: POTRA1-POTRA2(P1-2) and POTRA3-POTRA4-POTRA5(P3-5). Thus, for further data analysis, we analyzed BamA as four layers: the His6 extra-cellular layer where BamA is attached to the chip, the membrane layer (containing the β -barrel domain of BamA), an adjoining P3-5 layer, and the N-terminal P1-2 layer. NR data can then highlight variations in thickness of these layers, which corresponds to the position of additional protein mass in each point of measurement (**Fig. 5C** [↗](#), **fig. S10B, C**).

On addition of BamD (2nd measurement), the scattering length density (SLD) profiles showed that BamD was located in the P3-5 layer (**Fig. 5C** [↗](#), **fig. S10D, E**), an observation consistent with previous studies on the resting BAM complex ([Ding et al., 2020](#) [↗](#)). On addition of the OmpC(Y286A) substrate (3rd measurement) binding was observed to the P3-5 layer, with a

concomitant extension of this layer from 29.7 ± 0.8 Å to 45.5 ± 1.6 Å away from the membrane (Fig. 5C, 5D, fig. S10F,G). By contrast, no obvious changes were observed on the addition of the native OmpC protein implying that interactions it makes with BamA or BamD on the Si-Wafer were resolved too quickly to capture (fig. S11). The failure to observe changes with the WT-OmpC control in the NR analysis is consistent with the data from the BN-PAGE analysis (Fig. 4A) where native OmpC does not form the stable intermediate seen for OmpC(Y286A). The NR findings indicate that OmpC(Y286A) is stacked at a position in the BAM complex with the POTRA3-5 of BamA and BamD, and that the Y286A mutation results in a relatively stable binding to this specific periplasmic region of the BAM complex.

BamD acts as a receptor for the internal signal and β -signal

To elucidate the residues within BamD that mediate this signal-receptor interaction, we applied *in vitro* site-specific photo-crosslinking (fig. S12A). A library of 40 BamD variants was created to incorporate the non-natural amino acid, *p*-benzoyl-L-phenylalanine (BPA), by the suppressor tRNA method in a series of positions (Chin et al., 2002). We purified 40 different BPA variants of BamD, and then irradiated UV after incubating with 35 S-labelled OmpC. The UV-dependent appearance of crosslinks between OmpC and BamD occurred for 17 of the BamD variants (fig. S12B). Structurally, these amino acids locate both the lumen side of funnel-like structure (e.g. 49 or 62) and outside of funnel-like structure such as BamC binding site (e.g. 114) (fig. S12C). The analysis was repeated to determine BamD cross-links for 35 S-labelled OmpC(WT), the internal signal mutant OmpC(Y286A), or the β -signal mutant OmpC(β AAA: Y365A, Q366A, F367A) (fig. S12D). This mapping of OmpC binding showed that an N-terminal section of BamD (amino acid residues 60 or 65) interacts with the internal signal, while a distinct section of BamD (amino acid residues 196, 200, 204) interacts with β -signal (fig. S12E).

While *in vitro* site-specific photo-crosslinking provides information on the substrate receptor region of BamD, this data only indirectly measures which residues of OmpC are being recognized. As our peptidomimetic screen identified conserved features in the internal signal, and cross-linking highlighted the N-terminal and C-terminal TPR motifs of BamD as regions of interaction with OmpC, we focused on amino acids specifically within the β -signals of OmpC and regions of BamD which interact with β -signal. We purified a form of BamD that is substituted with a cysteine residue in the substrate-binding region, and synthesized forms of 35 S-labeled OmpC substituted with a cysteine residue at various positions. Treatment with 1 μ M CuSO₄ can stimulate disulfide formation between BamD and OmpC (Fig. 6A). Consistent with the BPA crosslinking data, the positions near the internal signal in OmpC (amino acid residues 278, 284, 296 and 302) formed disulfide bonds with the N-terminal section of BamD (amino acid residues 49, 65). Conversely, residues towards the C-terminal, canonical β -signal of OmpC (amino acid residues 333, 357, and 363) formed disulfide bonds with a C-terminal section of BamD (at amino acid residue 204) (Fig. 6B). All of disulfide crosslinked products were validated by reducing condition SDS-PAGE (fig. S13).

To further probe this interaction, we engineered an *E. coli* strain to express BPA-containing BamD and a FLAG-tagged form of OmpC and performed *in vivo* photo-crosslinking (Fig. 6C). These western blots reveal cross-linked products representing the interacting protein species. Photo cross-linking of unnatural amino acid is not a 100% efficient process, so the level of cross-linked products is only a small proportion of the molecules interacting in the assays. In BamD, BPA at positions 49, 53, 65, or 196 generated cross-links to FLAG-OmpC. Positions 114 and 200 in BamD were not detected to interact with OmpC. Positions 49, 53, 65, and 196 of BamD face the interior of the funnel-like structure of the periplasmic domain of the BAM complex, while position 114 is located outside of the funnel-like structure (Bakelar et al., 2016; Gu et al., 2016; Iadanza et al., 2016). We note that while position 114 was cross-linked with OmpC *in vitro* using purified BamD, that this was not seen with *in vivo* cross-linking. Instead, in the context of the BAM complex, position 114 of BamD binds to the BamC subunit and would not be available for substrate binding *in vivo* (Bakelar et al., 2016; Gu et al., 2016; Iadanza et al., 2016).

Two structurally distinct regions of BamD promote OmpC folding and assembly

In order to directly measure how the β -strands of OmpC come together, we established a new assay in which the folding of purified OmpC can be measured with or without assistance by purified BamD. Using the crystal structure of OmpC as a guide, cysteine residues were engineered in place of amino acids that were outward-facing on the final five β -strands of folded OmpC, placed close enough to a cysteine residue in the adjacent β -strand to allow for the possibility of disulfide formation once the strands had been correctly aligned in the folded protein (**Fig. 7A**). These cysteine-variant OmpC proteins were incubated with or without BamD to determine how efficiently disulfide bonds formed. Samples harboring cysteine residues in OmpC that were incubated with BamD migrated faster than non-Cys OmpC on non-reducing SDS-PAGE (**Fig. 7B**). This migration difference was not observed in reducing conditions, validating that the faster migration occurred as a result of internal molecule disulfide bond formation. Disulfide bonds were formed in each of the paired strands β -strands from -1 to -5, suggesting that BamD directly stimulates arrangement of C-terminal five strands of BamD.

Cross-linking studies shown in **Fig. 6B** indicated 2 distinct regions of BamD as hot-spots for OmpC-binding. Anticipating that binding sites would be conserved; we analyzed sequence conservations and found BamD(Y62) and BamD(R197) as good candidates proximal to OmpC binding sites (**Fig. 7C**). R197 has previously been isolated as a suppressor mutation of a BamA temperature sensitive strain (Ricci et al., 2012). Each residue was located on the lumen side of the funnel-like structure in the EspP-BAM assembly intermediate structure (PDBID: 7TTC) (Doyle et al., 2022). Mutant strains expressing BamD(Y62A) and BamD(R197A) in a genetic background where the chromosomal bamD can be depleted by an arabinose-responsive promoter were tested (**Fig. 7D and fig. S14A,B,C**). Growth conditions were established wherein we could deplete endogenous BamD but not disturb cell growth (fig. S14D, fig. S15A). Isolated EMM from this time point showed that neither the BamD(Y62A) nor the BamD(R197A) mutants affected the steady-state protein levels of other components of the BAM complex (fig. S15B), nor did they impact the complex formation of the BAM complex (**Fig. 7E**, fig. S15C). The EMM assembly assay showed that the internal signal binding site was as important as the β -signal binding site to the overall assembly rates observed for OmpC (**Fig. 7F**), OmpF (fig. S15D), and LamB (fig. S15E). These results suggest that recognition of both the C-terminal β -signal and the internal signal by BamD is important for efficient protein assembly.

BamD is responsible for OMP assembly to retain outer membrane integrity

In *E. coli*, BamD function is essential for cell viability (Onufryk et al., 2005). As a result, in an *E. coli* strain where bamD expression is under the control of an arabinose-inducible promoter, in a control strain (**Fig. 8A**, “vec”) two rounds of dilution in restrictive growth media depletes the level of BamD protein and stops cell growth (**Fig. 8A**). The BamD(Y62A) mutant and the BamD(R197A) mutant supported viability (**Fig. 8A**, fig. S15A), allowing membrane fractions to be isolated from each mutant strain: the steady-state level of porins OmpA and OmpC/OmpF was reduced in the BamD(Y62A) and BamD(R197A) mutants (**Fig. 8B**). This was also true when the level of trimeric OmpF was assessed by native PAGE: less OmpF trimer had been assembled over the growth period relative to the steady-state level of the BAM complex (**Fig. 8C**). Thus, while these mutant forms of BamD support cell viability (**Fig. 8A**), they have deleterious effects on β -barrel protein assembly *in vivo*. In the case of the BamD(Y62A) strain, growth in the presence of vancomycin showed impaired outer membrane integrity (**Fig. 8D,8E**).

Vancomycin sensitivity is a classic assay to measure membrane integrity in Gram-negative bacteria such as *E. coli* because the drug vancomycin cannot permeate the outer membrane except if outer membrane integrity is breached by any means. Then, and only then, is sensitivity to

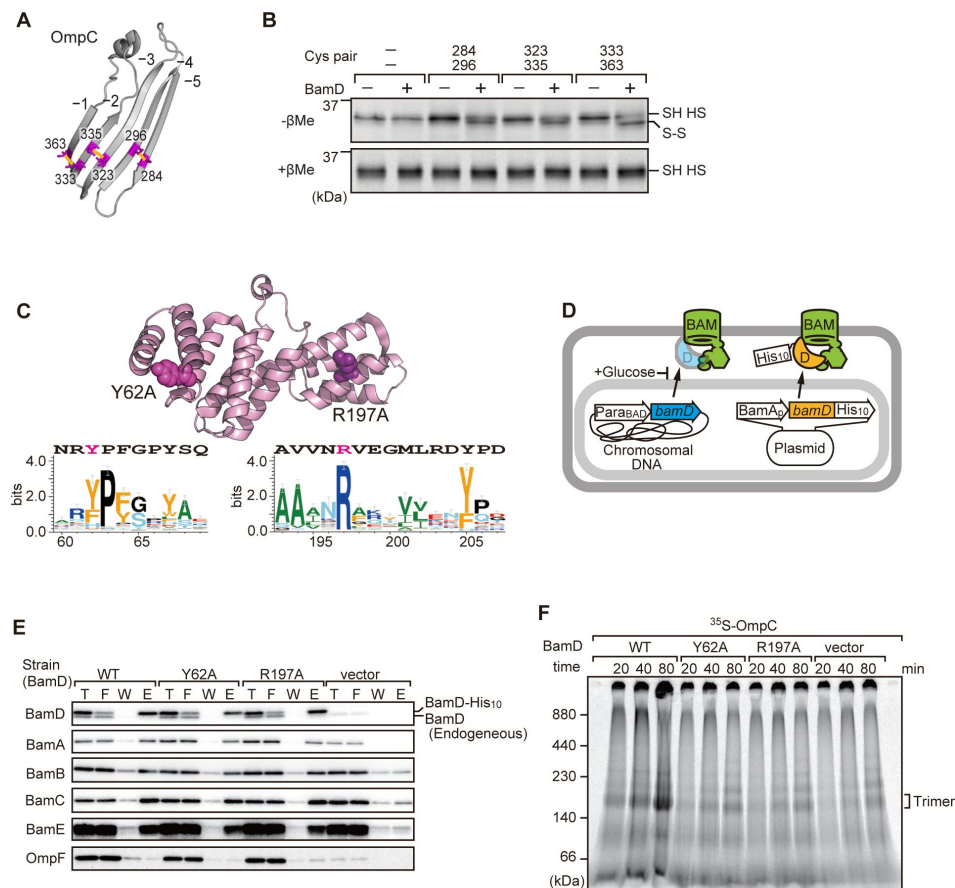


Fig. 7

Key residues in two structurally distinct regions of BamD promote β -strand formation and OMP assembly. **A**, Double cysteine residues were mutated into OmpC between anti-parallel β -strands to form artificial disulfide bonds **B**, ³⁵S-OmpC-cys variants were translated then incubated with (+) or without (-) BamD. The samples were analyzed by reducing or non-reducing condition. Disulfide bond specific bands (S-S) are indicated to the right of the gel. **C**, Sequence conservation analysis of BamD. Target residues, Y62 and R197, are indicated with dark pink spheres on the crystal structure of BamD, sequence logos represent conservation in the immediate region of these residues. Amino acids listed above logo sequence are from crystal structure. Sequence conservation of BamD (bottom). **D**, Schematic depiction of the BamD depletion strain of *E. coli* used to express mutant BamD proteins. **E**, Pull-down assay of the variants of BamD-His8. The EMMs as in **(C)** were solubilized with 1.5% DDM and then subjected to Ni-NTA. T- 5% total input, F- unbound fraction, W- wash fraction, E- Eluted fraction. Each fraction was analyzed by SDS-PAGE and immunoblotting against indicated antibodies. **F**, Assembly of OmpC was reduced in EMMs containing BamD mutated at Y62 and R197. ³⁵S-labelled OmpC was incubated with EMM as in **(C)**, and then analyzed by BN-PAGE and radio-imaging.

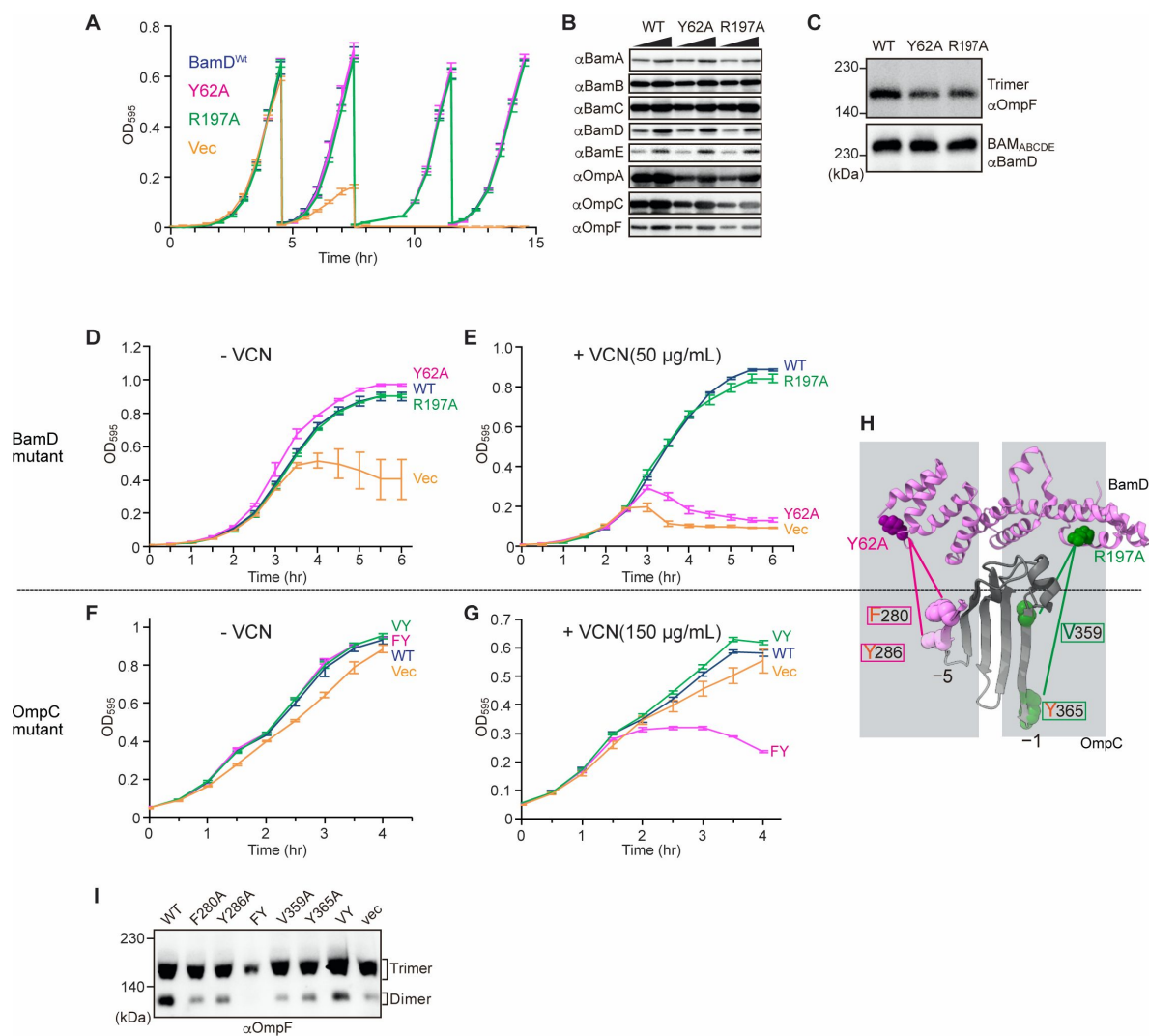


Fig. 8

Substrate recognition by BamD is essential to maintain membrane integrity. **A**,

Strains of *E. coli* with a depletion of chromosomal *bamD*, complemented by expression of the indicated forms of BamD, were grown in restrictive media (LB supplemented with 0.5% glucose) for 4 hours, then diluted into fresh media and grown a further 4 hours. **B**, Quantitation of steady-state protein levels of OmpF and the indicated subunits of the BAM complex in WT, Y62A, R197A mutants were determined after SDS-PAGE and immunoblot analysis. Wedge indicates that in each case, two samples were assessed corresponding to either 4 µg or 12 µg of total protein. **C**, Quantitation of the level of trimeric OmpF formed in the indicated EMM assays. The immunoblot analysis is shown after BN-PAGE analysis and immunoblotting with antibodies recognizing OmpF (upper panel) or BamD (lower panel). **D**, **E** *E. coli* *bamD* depletion cells expressing mutations at residues, Y62A and R197A, in the β-signal recognition regions of BamD were grown with or without VCN. **F**, **G**, *E. coli* cells expressing mutations to OmpC internal signal, as shown in Fig 3, grown in the presence of VCN. Mutations to two key residues of the internal signal were sensitive to the presence of VCN. **H**, Structure of BamD (top) and the final 5 β-strands of OmpC (bottom). Gray boxes indicate relative regions of interaction between BamD and substrates. **I**, OmpF complex assembly *in vivo* of cells expressing mutant OmpC. OmpF assembly was greatly reduced in cells expressing double mutations to the -5 internal β-signal.

vancomycin seen (Hart and Silhavy, 2020 [↗](#)). To test whether expression of a mutant form of OmpC that cannot engage BamD *in vivo* would likewise diminish membrane integrity, we assayed for growth of *E. coli* strains expressing OmpC(FY) and other mutant forms of OmpC (Fig. 8F [↗](#), 8G). The OmpC(FY) -5 strand double mutant showed increased sensitivity to vancomycin (Fig. 8G [↗](#)). The OmpC(FY) mutant does not accumulate in the outer membrane to the level found in wild-type *E. coli* (Fig. 3D [↗](#)). The strain co-expressing OmpC(FY) with endogenous OmpC has diminished levels of another major porin (OmpF) as judged by BN-PAGE, but not OmpC(WT) (Fig. 8I [↗](#), see also Fig. 3 [↗](#)). The dimeric form of endogenous OmpF was prominently observed in both the OmpC(WT) as well as the OmpC(VY) double mutant cells.

Discussion

OMPs are the major components of bacterial outer membranes, accounting for more than 50% of the mass of the outer membrane (Horne et al., 2020 [↗](#); Jarosławski et al., 2009 [↗](#)). The quantitative analysis of single-cell data suggests approximately 500 molecules of major porins, such as OmpC or OmpF, are assembled into the outer membrane per minute per cell (Benn et al., 2021 [↗](#); Lithgow et al., 2023 [↗](#)). This considerable task requires factors that promote the efficient and prompt assembly of OMPs in order to handle this extraordinary rate of substrate protein flux. The efficiency of assembly is necessary to fight against external cellular insult, hence the use of BamD as an OMP assembly accelerator.

In our experimental approach to assess for inhibitory peptides, specific segments of the major porin substrate OmpC were shown to interact with the BAM complex as peptidomimetic inhibitors. Results for this experimental approach went beyond expected outcomes by identifying the essential elements of the signal $\Phi\text{xxxxxx}[\Omega/\Phi]\text{x}[\Omega/\Phi]$ in β -strands other than the C-terminal strand. Discovery of conserved, internal signals in OMPs explains two previous reports where the -5 strand of BamA was suggested to be more similar to a β -signal motif than the C-terminal strand of BamA (Hagan et al., 2013 [↗](#); Imai et al., 2011 [↗](#)) and a hidden Markov model analysis that showed that in autotransporter OMPs, a class of OMPs which includes EspP, 473/1511 of non-redundant proteins show no β -signal motif at the C-terminal strand, but instead contained a β -signal motif in the -5 strand (Celik et al., 2012 [↗](#); Cox et al., 2010 [↗](#)).

The inhibitory peptides discovered here contained information that led us to define an internal signal for OMP assembly. This internal signal corresponds to the -5 strand in OmpC and is recognized by BamD. Sequence analysis shows that similar sequence signatures are present in other OMPs (Figs. S5, S6 and S7). These sequences were investigated in two further OMPs: OmpF and LamB (Fig. 2C and D [↗](#)). Hidden Markov model approaches like HMMER (Eddy, 1995 [↗](#)) detect conserved sequence features and tools such as MEME (Bailey et al., 2009 [↗](#)) can then define a sequence motif derived from information across very many non-redundant protein sequences. The abundance of information which arises from modeling approaches and from the multitude of candidate OMPs, is generally oversimplified when written as a primary structure description typical of the β -signal for bacterial OMPs (i.e. $\zeta\text{xGxx}[\Omega/\Phi]\text{x}[\Omega/\Phi]$) (Kutik et al., 2008 [↗](#)). Analysis of the few peptidomimetic peptide sequences in functional analyses, via both *in vitro* assays and intact *E. coli* analysis, suggests that $\Phi\text{xxxxxx}[\Omega/\Phi]\text{x}[\Omega/\Phi]$ is a descriptor for the internal signal, which is complementary in function to the C-terminal β -signal that engages the BAM complex to assist OMP assembly into the outer membrane.

The β -signal motif ($\zeta\text{xGxx}[\Omega/\Phi]\text{x}[\Omega/\Phi]$) is an eight-residue consensus, and internal signal motif is composed of a nine-residue consensus. Recent structures have shown the lateral gate of BamA interacts with a 7-residue span of substrate OMPs. Interestingly, inhibitory compounds, such as darobactin, mimic only three residues of the C-terminal side of β -signal motif. Cross-linking presented here in our study showed that BamD residues R49 and G65 cross-linked to the positions 0 and 6 of the internal signal in OmpC (Fig. 6D [↗](#)). Both signals are larger than the assembly

machineries signal binding pocket, implying that the signal might sit beyond the bounds of the signal binding pocket in BamD and the lateral gate in BamA. These findings are consistent with similar observations in other signal sequence recognition events, such as the mitochondrial targeting presequence signal that is longer than the receptor groove formed by the Tom20, the subunit of the translocator of outer membrane (TOM) complex (Yamamoto et al., 2011 [DOI](#)). The presequence has been shown to bind to Tom20 in several different conformations within the receptor groove (Nyirenda et al., 2013 [DOI](#)).

The current dogma in our field states that the information contained in the β -signal is important for engagement into the lateral gate of BamA (Bitto and McKay, 2003 [DOI](#); de Cock et al., 1997 [DOI](#); Doyle et al., 2022 [DOI](#); Doyle and Bernstein, 2019 [DOI](#); Hagan et al., 2015 [DOI](#); Jansen et al., 2000 [DOI](#); Robert et al., 2006 [DOI](#); Tomasek et al., 2020 [DOI](#); Xiao et al., 2021 [DOI](#)). This interaction initiates the insertion of a nascent OMP polypeptide into the outer membrane. By this model, BamD is considered to function by organizing the BamC and BamE subunits of the BAM complex against the core subunit BamA (Kim et al., 2011 [DOI](#), 2007 [DOI](#); Malinverni et al., 2006 [DOI](#)). If that is the sole role of BamD, then BamD is essential because its importance in structural organization of the BAM complex (Hart and Silhavy, 2020 [DOI](#)), not because of a substrate recognition function. Previous studies have shown BamD is capable of stimulating the folding and insertion of OMPs into liposomes in the absence of BamA (Hagan et al., 2013 [DOI](#), 2010 [DOI](#)). The question remains, how is BamD capable of this function? In our present study, we show that incubation of OmpC with BamD promotes the formation of antiparallel β -strands by the formation of intra-molecular disulfide cross-linking of neighboring β -strands at the C-terminus (**Fig. 7B** [DOI](#)). In EspP position 1214, located within the internal signal, was seen to cross-link to BamD (Ieva et al., 2011 [DOI](#)), and our inter-molecular cysteine cross-linking showed the N-terminal helices of BamD's TPR structure interacts with the internal signal and the C-terminal helices interacts with the β -signal (**Fig. 6B, C** [DOI](#)). Failure in this interaction was seen to stall substrate assembly within the soluble domain of the BAM complex (**Fig 4C** [DOI](#) and **5D** [DOI](#)). This suggests that BamD acts in an important role in an early recognition step in which both the substrate's C-terminal β -signal and internal signal engage BamD in order to be arranged into a form ready for transfer to the lateral gate of BamA (Shen et al., 2023 [DOI](#); Tomasek et al., 2020 [DOI](#)) (**Fig 9A** [DOI](#)). This adds an additional role of the C-terminus of BamD beyond a complex stability role (Ricci et al., 2012 [DOI](#); Storek et al., 2023 [DOI](#)). The recognition and initiation of β -strand formation by BamD is necessary to support the efficient assembly of the 500 molecules of OMPs per minute required by the outer membrane, ensuring the survival against foreign insult and supports bacterial survival (Lithgow et al., 2023 [DOI](#)).

Recent advances in our understanding of BAM complex function have come from structural determination of EspP engaged in assembly by the BAM complex (Doyle et al., 2022 [DOI](#); Shen et al., 2023 [DOI](#)). The assembly intermediate of EspP engaged with the BAM complex shows the -4 strand, interacts with the R49 residue of BamD, in the N-terminal substrate binding region, even before the -5 strand had entered into this cavity (Doyle et al., 2022 [DOI](#)). Spatially, this indicates the BamD can serve to organize two distinct parts of the nascent OMP substrate at the periplasmic face of the BAM complex, either prior to or in concert with, engagement to the lateral gate of BamA. Assessing this structurally showed the N-terminal region of BamD (interacting with the POTRA1-2 region of BamA) and the C-terminal region of BamD (interacting with POTRA5 proximal to the lateral gate of BamA) (Bakelar et al., 2016 [DOI](#); Gu et al., 2016 [DOI](#); Tomasek et al., 2020 [DOI](#)) has the N-terminal region of BamD changing conformation depending on the folding states of the last four β -strands of the substrate OMP, EspP (Doyle et al., 2022 [DOI](#)). The overall effect of this being a change in the dimensions of this cavity change, a change which is dependent on the folded state of the substrate engaged in it (**Fig 9** [DOI](#) B-E). We showed that purified BamD can facilitate partial folding of an OMP and we propose that BamD captures the -1 strand and draws the -5 strand by a conformational change, thereby catalyzing the formation of hydrogen bonds to form β -hairpins in neighboring strands, to prime the substrate for insertion into the outer membrane by the action of BamA. After OMP assembly, all elements of the internal signal are positioned such that they face into the lipid-phase of the membrane. This observation may be a coincidence, or may be utilized by the BAM

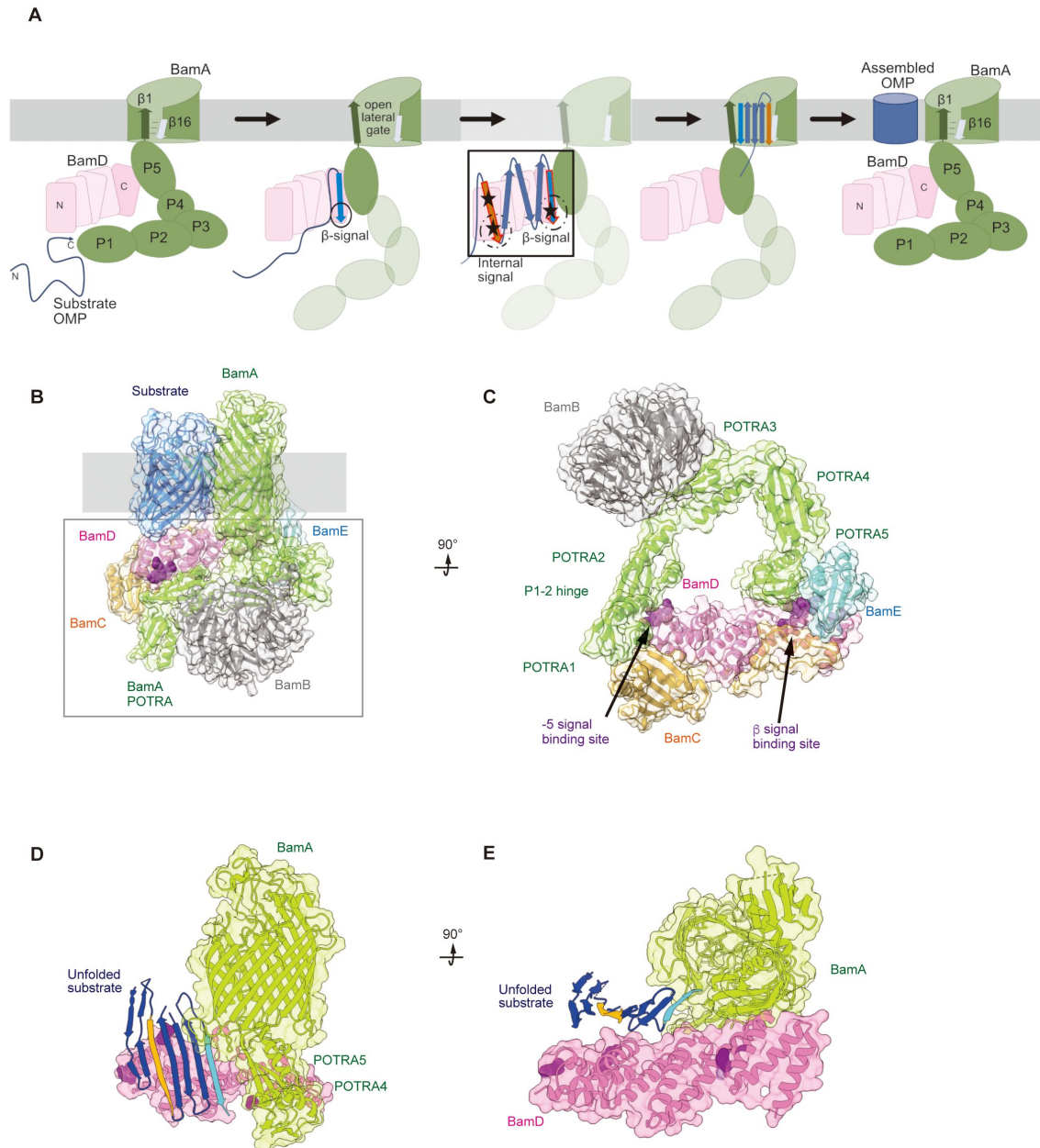


Fig. 9

Model for the recognition of internal and canonical β -signals by the BAM complex. **A**, BAM complex assisted-OMP assembly is initiated when substrate OMPs encounter the periplasmic domain of the BAM complex. The canonical β -signal (blue arrow) of the OMP engages with the C-terminal TPR motifs of BamD (pink), after which the internal β -signal (orange arrow) is marshalled by the N-terminal TPR motif. The internal signal interacts with BamD R49 and N60 (black stars) while the canonical β -signal interacts with BamD D204 (black star). The proximity of BamD-substrate to POTRA 5 and the organization of the C-terminal β -strands of OMPs can then stimulate the entrance to the lateral gate of BamA. Finally, barrel formation and release into the membrane follows as previously published^{28,29}. **B**, Structure of substrate engaged BAM complex (PDBID: 6V05). **C**, Bottom-up view of periplasmic domain of complex as shown in (B). Arrows indicate the position of the residues on BamD that function in binding sites for -5 signal and the β signal. **D**, A cavity formed by POTRA 1-2 and POTRA 5 of BamA and BamD is sufficient to accommodate the C-terminal five β -strands of substrate OMP. Substrate -1 strand and -5 β -strand are indicated by light blue strand and orange strand, respectively. The signal binding region of BamD is emphasized by deep purple (PDBID: Bam complex: 7TT5 and 7TT2). **E**, Bottom-up view of complex as shown in (D).

complex to register and orientate the lipid facing amino acids in the assembling OMP away from the formative lumen of the OMP. Amino acids at position 6, such as Y286 in OmpC, are not only component of the internal signal for binding by the BAM complex, but also act in structural capacity to register the aromatic girdle for optimal stability of the OMP in the membrane.

Materials and methods

E. coli strains and Growth conditions

E. coli strains used in this study are listed in Table S1. All strains were grown in LB (1% tryptone, 1% NaCl, and 0.5% yeast extract). *E. coli* strains used for the *in vivo* photo-crosslinking experiments were grown in XB (1% tryptone, 1% NaCl, and 0.1% yeast extract). Unless otherwise specified, ampicillin (amp; 50 mg/ml), chloramphenicol (Cm; 10 mg/ml), or kanamycin (Kan, 30 mg/ml) were added to media for plasmid selection.

To create the BamD-depletion strain, we amplified four DNA fragments encoding 540 to 40 base pairs of the 5' UTR of BamD (primers BamDSO-1 and -2), kanamycin cassette (primers BamDSO-3 and -4), AraC-pBAD (primers BamDSO-5 and -6), and the first 500 bp of BamD (primers BamSO-7 and -8), respectively (Figure S16). Table S2 describes the pair of primers and template DNA to amplify each DNA fragment. These four DNA fragments were combined by the overlap PCR method and transformed into MC4100a harboring pKD46 encoding λ Red recombinase (Datsenko and Wanner, 2000 [\[4\]](#)). Transformants were selected in the presence of kanamycin and 0.1% (w/v) L -arabinose.

To culture BamD-depletion strains a single colony from each BamD-depletion strains harboring pBamD variants was inoculated into LB (+ amp + Kan + 0.02% (w/v) L -arabinose) and cultured for 10 h at 37°C. The saturation culture was shifted into the LB (+ amp + Kan) and incubated for another 10h at 37°C. The saturation culture was diluted into LB (+ amp + kan + 0.5% (w/v) D -Glucose) to an $\text{OD}_{600} = 0.02$, and incubated at 37°C until cell density reached $\text{OD}_{600}=0.8$. A second dilution in the same media to an $\text{OD}_{600} = 0.02$ was performed, and incubated for 2.5h at 37°C.

Plasmids

Plasmids and primers used in this study are listed in Tables S3 to S7. Cloning of specific target gene ORFs into specified vectors was performed via SLiCE *in vitro* recombination (Okegawa and Motohashi, 2015 [\[4\]](#)).

Protein Purification

Plasmids encoding each subunit of the BAM complex and OmpC genes (including a His₆-tag) were transformed into BL21(DE3)* cells (Invitrogen). BamD and OmpC variant transformants were grown in LB (+ amp) at 37°C to an $\text{OD}_{600} = 0.5$. Expression was induced by adding 1 mM IPTG and the cells grown at 37°C for 3 h. For BamD containing BPA mutations, MC4100 cells were transformed with pHIS-BamD-(X)-amber and pEVOL-BpF45. Transformants were cultured in LB with antibiotics to an $\text{OD}_{600} = 0.5$, then supplemented with 1 mM BPA, 0.05% (w/v) L -arabinose and 1 mM IPTG to induce expression. Cells were grown at 37°C for a further 3 h.

Cells were pelleted by centrifugation at 6,000 x *g* for 10 min. Pellets were resuspended in Buffer A (150 mM NaCl, 20 mM NaPO₄) and lysed via sonication. Lysates were clarified by low-speed centrifugation of 1100 x *g* for 10 min at 4°C. For purification of BamD variant proteins, supernatants were again clarified by high-speed centrifugation, 17000 x *g* for 15 min at 4°C. High-speed clarified supernatants were then applied Ni₂⁺-NTA-affinity column chromatography. Ni₂⁺-NTA resin were washed by Buffer A with 50 mM imidazole and proteins were eluted in Buffer A containing an imidazole step gradient (100 - 400 mM). Peak fractions were collected as the purified

fraction. To purify OmpC, the pellets containing OmpC in inclusion bodies were resuspended in Buffer A containing 0.5% Triton X-100 and centrifuged for 11,600 x *g* for 10 min at 4°C. Inclusion bodies were denatured in Buffer A with 6 M urea and mixed at 25°C for 90 min, followed by centrifugation of 11,600 x *g* for 10 min at 4°C. The filtered supernatant was applied to Ni₂⁺-NTA-affinity column chromatography. The bound OmpC-His₆ was washed with Buffer A (+ 6 M urea, 50 mM imidazole) and eluted with Buffer A containing 6M urea and 200 mM imidazole. Purified proteins were then diluted with glycerol (final concentration of 10%), snap frozen in liquid nitrogen and stored in - 80°C. BamA was purified as detailed previously (Ding et al., 2020 [DOI](#)).

Isolation of the *E.coli* microsomal membrane fraction (EMM)

E. coli crude membrane fraction were isolated as previously described (Gunasinghe et al., 2018 [DOI](#)). In brief, pelleted cells were resuspended in sonication buffer (50 mM Tris- HCl pH 7.5, 150 mM NaCl, 5 mM EDTA) and lysed via sonication on ice. After low- speed centrifugation (1,100 x *g*, 5 min, 4°C), supernatant was centrifuged again (15,000 x *g*, 10 min, 4°C). Pelleted membranes were resuspended in SEM buffer (250 mM sucrose, 10 mM MOPS-KOH pH 7.2, 1 mM EDTA) and flash frozen. Protein concentrations of EMM were calculated by OD₂₈₀ measurements of 100 times diluted in 0.6% SDS (OD₂₈₀=0.21 was set to be 10 mg/ml).

EMM assembly assay

The EMM assembly assay has been previously described (Gunasinghe et al., 2018 [DOI](#); Thewasano et al., 2023 [DOI](#)). The method in brief is as follows; substrates were prepared via *in vitro* transcription by SP6-RNA polymerase followed by *in vitro* translation in rabbit reticulocyte lysate supplemented with ³⁵S-methionine. EMMs were resuspended in Assembly Assay buffer (10 mM MOPS-KOH pH7.2, 2.5 mM KH₂PO₄, 250 mM sucrose, 15 mM KCl, 5 mM MgCl₂, 2 mM methionine, 5 mM DTT, 1% w/v BSA, 0.09% v/v Triton X-100), and incubated with ³⁵S-labelled substrate at 30°C for indicated time points. Assembly reactions were halted by moving to ice for 5 minutes. The EMMs were then harvested via centrifugation at 15,000 x *g* for 5 min at 4°C and washed with SEM buffer. After SEM buffer wash, trimeric protein assembled EMM pellets were solubilized in 1.5% DDM containing Blue native (BN)-PAGE Lysis Buffer (25 mM imidazole-HCl pH 7.0, 50 mM NaCl, 50 mM 6-aminohexanoic acid, 1 mM EDTA, 7.5% (w/v) glycerol) on ice for 20 minutes. Solubilized proteins were clarified by centrifugation of 15,000 x *g* for 10 min at 4°C, followed by BN-PAGE analysis as described previously (Gunasinghe et al., 2018 [DOI](#); Thewasano et al., 2023 [DOI](#)).

Peptide library was synthesized by MIMOTOPES, and resuspended in DMSO at a concentration of 30 mM. The peptide inhibition screen of EspP was performed as stated above, but with the addition of 0.1 mM inhibitory peptide to the Assembly Assay buffer and incubated with EMMs for 5 minutes prior to the addition of ³⁵S-labelled substrate protein. After harvesting, EMMs were treated with 100 µg/mL Proteinase K on ice for 20 minutes. Digestion was halted by adding 2 mM PMSF. The EMMs were then collected via centrifugation at 15,000 x *g* for 5 min at 4°C and washed with SEM buffer. EspP assembled with EMM proteins were analyzed by SDS-PAGE and radio-imaging. EMM assembly of trimeric proteins was tested with the addition of 0.25 mM peptide and analyzed as above.

Sequence conservation analysis

For conservation analysis, we obtained homologous sequences of representative structure-known beta barrel outer membrane proteins from reference bacterial proteomes in UniProt (The UniProt Consortium, 2019) using phmmer and jackhammer HMMER 3.2.1 (<http://hmmer.org> [DOI](#)). The best hit sequences in organisms which also contained a BamD homolog were used. We identified 31, 94, 25, 19, 27, 44, 43, 23, 42, 22, 132, 46, 58, 68, 39, 245, 261, 115, 187, 250 and 74 sequences for OmpA, OmpW, PagP, OmpX, OmpT, EspP, Hbp, NanC, OMPLA, Tsx, FadL, OmpC, OmpF, PhoE, LamB, BtuB, Cir, FecA, FepA, FhuA and PapC, respectively (sequence identity less than 60%, expect for OmpX. In

case of OmpX, sequence identity is less than 80% due to the small number of sequences). Multiple alignments were generated by MAFFT (Kato et al., 2019) and then sequence logos of -5th transmembrane strands were created using WebLogo 3 (Crooks, 2004).

In vivo assembly analysis of FLAG-OmpC

We introduced a FLAG epitope tag into the N-terminal region of OmpC, immediately behind the cleavage site for the SEC-signal sequence (Rapoport, 2007), to distinguish between endogenous OmpC and the OmpC variants in the OmpC-deletion strain background which has issues of diminished OM integrity. The expression of FLAG- OmpC mutants was controlled by using a plasmid-based pBADara promoter cassette, to avoid the chronic effect of constitutively expressed mutant proteins (Figure 3A). We transformed pBAD-FOmpC variants into MC4100A. Transformants were cultured in LB (+ amp) until OD600=0.5 and then protein expression induced by adding 0.1% arabinose for 3 h. *E. coli* cells were harvested and total cell lysate or EMMs prepared accordingly.

Antibody Shift Assay

The variation of BN-PAGE analysis referred to as an antibody shift assay was performed as previously described with modification (Shiota et al., 2012). In brief, solubilized EMMs were incubated with 3 μ L of anti-BamC antibodies for 45 min at 4°C. Samples were clarified via centrifugation at 13,000 x *g* for 10 minutes at 4°C, and subjected to BN-PAGE.

Urea Extraction of Intermediate

Once the Assembly Assay was performed, to obtain intermediate complexes, lysates were washed with SEM, the EMMs resuspended in SEM containing 6 M urea and incubated at 37°C for 1 hour, after which 50 mM Tris HCL pH 8.0 was added. Membrane fractions were collected via high-speed centrifugation (100,000 x *g*, 45 min, 25°C) and washed twice with SEM buffer. The membrane was solubilized for 20 minutes at 4°C in 1.5% DDM containing BN-PAGE lysis buffer and subjected to BN-PAGE analysis.

Neutron Reflectometry

Neutron reflection (NR) was carried out on the SOFIA reflectometer at the Materials and Life Science Experimental Facility (MLF) at J-PARC, Japan (Mitamura et al., 2013; Yamada et al., 2011). The reflectivity at the interface between the Si substrate and water solution was measured at 3 incident angles (0.3°, 0.75° and 1.8°). The neutrons were introduced from the substrate side to illuminate the interface and the reflection intensity was normalized by the transmission intensity through the substrate to achieve the reflectivity with the *Q* range of up to 0.2 Å⁻¹. Neutron reflectivity, denoted as *R*, is the ratio of the incoming to the exiting neutron beams. It's calculated based on the Momentum transfer *Q*, which is defined by the formula $Q=4\pi \sin\theta/\lambda$, where θ represents the angle of incidence and λ stands for the neutron wavelength. The approximate value of *R(Q)* can be expressed as $R(Q)=16\pi^2/Q^2 |\rho(Q)|^2$, where *R(Q)* is the one-dimensional Fourier transform of $\rho(z)$, which is the scattering length density (SLD) distribution perpendicular to the surface. SLD is calculated by dividing the sum of the coherent neutron scattering lengths of all atoms in a sample layer by the volume of that layer. Consequently, factors such as thickness, volume fraction, and interface roughness of the samples significantly influence the intensity of the reflected beams.

Analysis of the NR profiles was performed using the MOTOFIT analysis software (Nelson, 2010). The software uses a conventional method, optical matrix method (Nelson, 2006) to model the reflectivity profiles. Briefly, the layer is divided into several sublayers and then a characteristic matrix is evaluated for each sublayer, from which the whole reflectivity can be calculated exactly. The same layer with multiple datasets with three different isotopic contrasts is fitted simultaneously until the satisfactory fit is obtained (smallest χ^2 value in the genetic algorithm). All

models include a 15 Å oxide layer on the surface of the silicon substrates and subsequently there are three layers of Ni, gold and Ni-NTA on the top of oxide layer. Protein layers will assemble outwards from the Ni-NTA layer.

The best-fit model from MOTOFIT gives information about the thickness, density and roughness of each layer. When a layer is composed of just two components, a chemical species or its fragments and water w , the SLD is given by: where ρ_w and ρ_s are the scattering length densities of the two components, respectively and ϕ is the volume fraction of species s in the layer. If more than one chemical species is present, then ρ_s is determined by the volume fraction of each species in the layer.

OmpC *in vitro* folding

Urea denatured OmpC and OmpC variants were purified as stated above. Protein was resuspended in Refolding Buffer (10 mM Phosphate Buffer pH 8.0, 2 mM EDTA, 1 mM glycine) to a final concentration of 5 μ M. DDM was added to a final concentration of 3.4 mM. Samples were incubated at 37°C for 0, 1, 3, 6, 8, 12, 24, and 36 hours. SDS-Loading dye was added and each sample was split into two tubes (one to be boiled, one at RT for 10min). OmpC folding was analyzed via SDS-PAGE.

Ni-NTA Substrate Pull-down

Purified BamA or BamD was resuspended in Buffer A to a final concentration of 10 μ M. Specific peptides or DMSO alone was added (final concentration 0.35 mM) and incubated rocking at 4°C for 1 hour. To each tube, 0.5 μ L of 35 S-OmpC was added and vortexed vigorously, 10% of the total volume was removed to assay as “input” for each sample. Samples were then incubated at 4°C with rocking for 30 minutes. Ni-NTA agarose beads were added to each sample, incubated at 4°C with rocking for an additional 30 minutes then washed with Buffer A containing 20 mM imidazole. BAM proteins were eluted from the Ni-NTA resin using Buffer A with 400 mM imidazole. Proteins were TCA precipitated and subjected to analysis by SDS-PAGE and imaged via radio-imaging.

Disulfide cross-linking

BamD-OmpC cysteine crosslinking was performed using 50 μ M purified single-cysteine BamD mutant variants incubated with 8 μ L of single-cysteine OmpC variants translated by the retic in 30 μ L of Buffer A at 25°C for 2 min. An oxidizing reaction mixture of 1 μ M CuSO₄, 4 mM methionine, and 4 mM cysteine was added and incubated at 25°C for 5 min. Proteins were precipitated by TCA precipitation, and solubilized in 1% SDS buffer. Solubilized proteins were diluted by 0.5% TritonX-100 buffer and purified with Ni-NTA via the tagged BamD-H6. BamD and disulfide crosslink products were eluted with 400 mM imidazole containing 0.5% TritonX-100 buffer. Eluted proteins were concentrated by TCA precipitation and solubilized with SDS-PAGE loading dye with or without β -Me.

The OmpC intrachain disulfide bond assay was performed using 1 μ L 35 S-OmpC double cysteine variants translated in the presence of 1 mM DTT. OmpC variants were diluted into 100 μ L Buffer A then incubated with 50 μ M BamD at 32°C for 30 minutes. 10 μ M CuSO₄ was added to samples and incubated for an additional 2 minutes, followed by the addition of SDS-PAGE loading dye with or without 5mM DTT and 1% β -Me.

Proteins were resolved via SDS-PAGE and imaged as stated above.

In vivo and *in vitro* photo-crosslinking

For *in vivo* photo-crosslinking, MC4100 strain was transformed with pEVOL-pBpF, pTnT-BamAp-BamD-His8, and pET-FLAG-OmpC, with respective antibiotics. A single colony was used to inoculate a culture in photo-crosslinking media (1% NaCl, 1% Tryptone, 0.1% Extract yeast dried,

1mM BPA) for 3 hours at 37°C in the dark.

Afterward, arabinose (0.1% final concentration) was added and further incubated for 2.5 h. Cultures were divided with one half UV-irradiated for 7 min at RT and the other half not. Cells were pelleted and solubilized with 1% SDS buffer (50mM Tris-HCl pH 8.0, 150mM NaCl, 1% SDS). His-tagged BamD containing BPA and its cross-linked products were purified by Ni-NTA agarose. Cross-linked products were identified via immunoblotting with anti-BamD and anti-FLAG antibodies.

In vitro photo-crosslinking was performed with 32 µg of purified BamD-His₆ variants containing BPA diluted in Buffer A. 40 µL of ³⁵S-OmpC was added to each tube and incubated at 25°C for 10 minutes in the dark. Each mixture was transferred to the lid of a microtube and irradiated with UV using B-100AP (UVP) for 7 min on ice. 10% (w/v) SDS solution was added to a final concentration of 1%, then incubated at 95°C for 10 min to denature proteins. Protein mixtures were diluted with 0.5% TritonX-100 buffer (50 mM Tris-HCl pH 7.0, 150 mM NaCl, 0.5% Triton-X100), and purified on Ni-NTA agarose beads. Elute fractions were concentrated by TCA precipitation and analyzed by SDS- PAGE and radio-imaging.

Statistical analysis

For all densitometry measurements we utilized “imagequant” quant (ImageQuant TL, cytiva) and statistical analysis and graphing was performed with Microsoft Excel software.

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Competing interests

Authors declare that they have no competing interests.

Data and materials availability

All data are available in the main text or the supplementary materials.

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Joint Public Review:

The biogenesis of outer membrane proteins (OMPs) into the outer membranes of Gram-negative bacteria is not fully understood, particularly client recognition and insertion by the conserved beta-assembly machinery (BAM), which is itself integrated in the outer membranes. So far, the last strand of an OMP, referred to as the beta-signal, has been known as a primary recognition motif by BAM. Here, authors have identified additional sequence motifs that are located in the upstream of the last strand.

Here, authors carried out rigorous biochemical, biophysical, and genetic approaches to prove that the newly identified internal motifs are critical to the assembly of outer membrane proteins as well as to the interaction with the BAM complex. The identification of important regions on the substrates and Bam proteins during biogenesis is an important contribution that gives clues to the path substrates take en route to the membrane. Assessing the effect of the internal motifs in the assembly of model OMPs in the absence (in vitro) and presence (in

vitro and in vivo) of BAM machinery aids a precise definition of the role of the motifs, solidifying the conclusions.

The initial reviews raised several concerns:

1. Strengthening the claim that the recognition of the internal signal by BAM is mediated by BamA and BamD via specific interactions.
2. Justification of the rationale of the peptide inhibition assays as a primary tool to identify significant recognition motifs.
3. More careful interpretation of the mutational effects on OMP assembly - that is, discerning the impairment of BAM-nascent polypeptide chain interaction from the impairment of intrinsic folding.
4. Providing further clarification of the discrepancy between in vitro assay and in vivo assay regarding the effect of the mutation Y286A on OMP assembly.
5. More elaboration on the rationale, interpretation, and representation of neutron refractory data.
6. An explanation is lacking why the strain with BamD R197A does not display VCN sensitivity in contrast to the strain with BamD Y62A.

Those concerns were well addressed in the revised manuscript in a rigorous manner.

Overall, this study comprehensively addresses an important question in the field. The notion that additional signals assist in biogenesis is a novel concept that has been tested and verified at least for a subset of model OMPs in this study. The generalization of the conclusion awaits a further proof of the concept.

- <https://doi.org/10.7554/eLife.90274.2.sa0>

Author Response

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

The biogenesis of outer membrane proteins (OMPs) into the outer membranes of Gram-negative bacteria is still not fully understood, particularly substrate recognition and insertion by beta-assembly machinery (BAM). In the studies, the authors present their studies that in addition to recognition by the last strand of an OMP, sometimes referred to as the beta-signal, an additional signal upstream of the last strand is also important for OMP biogenesis.

Strengths:

1. Overall the manuscript is well organized and written, and addresses an important question in the field. The idea that BAM recognizes multiple signals on OMPs has been presented previously, however, it was not fully tested.
1. The authors here re-address this idea and propose that it is a more general mechanism used by BAM for OMP biogenesis.

1. *The notion that additional signals assist in biogenesis is an important concept that indeed needs fully tested in OMP biogenesis.*

1. *A significant study was performed with extensive experiments reported in an attempt to address this important question in the field.*

1. *The identification of important crosslinks and regions of substrates and Bam proteins that interact during biogenesis is an important contribution that gives clues to the path substrates take en route to the membrane.*

Weaknesses:

Major critiques (in no particular order):

1. *The title indicates 'simultaneous recognition', however no experiments were presented that test the order of interactions during OMP biogenesis.*

We have replaced the word “Simultaneous” with “Dual” so as not to reflect on the timing of the recognition events for the distinct C-terminal signal and -5 signal.

1. *Aspects of the study focus on the peptides that appear to inhibit OmpC assembly, but should also include an analysis of the peptides that do not to determine this the motif(s) present still or not.*

We thank the reviewer for this comment. Our study focuses on the peptides which exhibited an inhibitory effect in order to elucidate further interactions between the BAM complex and substrate proteins, especially in early stage of the assembly process. In the case of peptide 9, which contains all of our proposed elements but did not have an inhibitory effect, there is the presence of an arginine residue at the polar residue next to hydrophobic residue in position 0 (0 Φ). As seen in Fig S5, S6, and S7, there are no positively charged amino acids in the polar residue positions in the -5 or last strands. This might be the reason why peptide 9, as well as peptide 24, the β-signal derived from the mitochondrial OMP Tom40 and contains a lysine at the polar position, did not display an inhibitory effect. Incorporating the reviewer's suggestions might elucidate conditions that should not be added to the elements, but this is not the focus of this paper and was not discussed to avoid complicating the paper.

1. *The β-signal is known to form a β-strand, therefore it is unclear why the authors did not choose to chop OmpC up according to its strands, rather than by a fixed peptide size. What was the rationale for how the peptide lengths were chosen since many of them partially overlap known strands, and only partially (2 residues) overlap each other? It may not be too surprising that most of the inhibitory peptides consist of full strands (#4, 10, 21, 23).*

A simple scan of known β-strands would have been an alternative approach, however this comes with the bias of limiting the experiments to predicted substrate (strand) sequences, and it presupposes that the secondary structure element would be formed by this tightly truncated peptide.

Instead, we allowed for the possibility that OMPs meet the BAM complex in an unfolded or partially folded state, and that the secondary structure (β-strand) might only form via β-argumentation after the substrate is placed in the context of the lateral gate. We therefore used peptides that mapped right across the entirety of OmpC, with a two amino acid overlap.

To clarify this important point regarding the unbiased nature of our screen, we have revised the text:

(Lines 147-151) "We used peptides that mapped the entirety of OmpC, with a two amino acid overlap. This we considered preferable to peptides that were restricted by structural features, such as β -strands, in consideration that β -strand formation may or may not have occurred in early-stage interactions at the BAM complex."

1. It would be good to have an idea of the propensity of the chosen peptides to form β -stands and participate in β -augmentation. We know from previous studies with darobactin and other peptides that they can inhibit OMP assembly by competing with substrates.

We appreciate the reviewer's suggestion. However, we have not conducted biophysical characterizations of the peptides to calculate the propensity of each peptide to form β -stands and participate in β -augmentation. The sort of detailed biophysical analysis done for Darobactin (by the Maier and Hiller groups, The antibiotic darobactin mimics a β -strand to inhibit outer membrane insertase Nature 593:125-129) was a Nature publication based on this single peptide. A further biophysical analysis of all of the peptides presented here goes well beyond the scope of our study.

1. The recognition motifs that the authors present span up to 9 residues which would suggest a relatively large binding surface, however, the structures of these regions are not large enough to accommodate these large peptides.

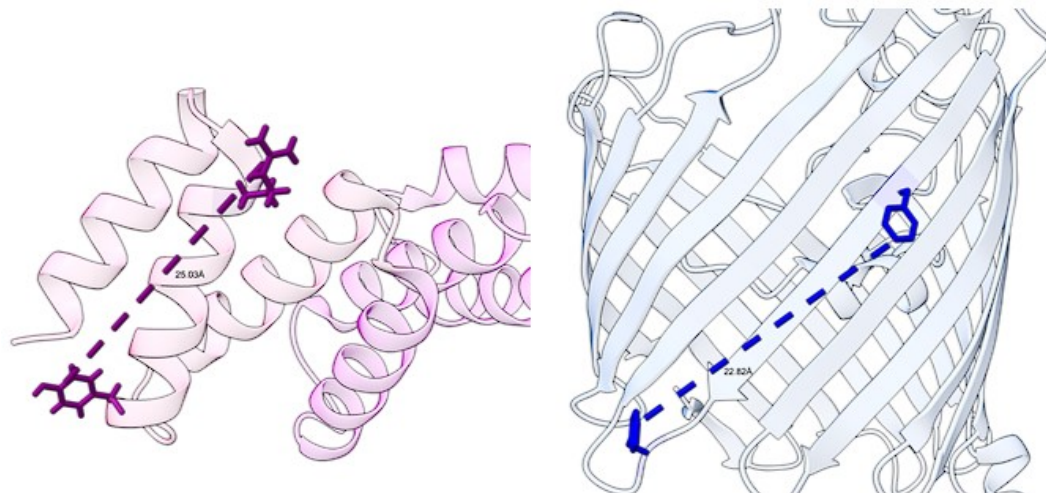
The β -signal motif (ζ xGxx[Q/ Φ]x[Q/ Φ]) is an 8-residue consensus, some of the inhibitory peptides include additional residues before and after the defined motif of 8 residues, and the lateral gate of BamA has been shown interact with a 7-residue span (eg. Doyle et al, 2022). Cross-linking presented in our study showed BamD residues R49 and G65 cross-linked to the positions 0 and 6 of the internal signal in OmpC (Fig. 6D).

We appreciate this point of clarification and have modified the text to acknowledge that in the final registering of the peptide with its binding protein, some parts of the peptide might sit beyond the bounds of the BamD receptor's binding pocket and the BamA lateral gate:

(Lines 458-471) "The β -signal motif (ζ xGxx[Q/ Φ]x[Q/ Φ]) is an eight-residue consensus, and internal signal motif is composed of a nine-residue consensus. Recent structures have shown the lateral gate of BamA interacts with a 7-residue span of substrate OMPs. Interestingly, inhibitory compounds, such as darobactin, mimic only three residues of the C-terminal side of β -signal motif. Cross-linking presented here in our study showed that BamD residues R49 and G65 cross-linked to the positions 0 and 6 of the internal signal in OmpC (Fig. 6D). Both signals are larger than the assembly machineries signal binding pocket, implying that the signal might sit beyond the bounds of the signal binding pocket in BamD and the lateral gate in BamA. These findings are consistent with similar observations in other signal sequence recognition events, such as the mitochondrial targeting presequence signal that is longer than the receptor groove formed by the Tom20, the subunit of the translocator of outer membrane (TOM) complex (Yamamoto et al., 2011). The presequence has been shown to bind to Tom20 in several different conformations within the receptor groove (Nyirenda et al., 2013)."

Moreover, the distance between amino acids of BamD which cross-linked to the internal signal, R49 and Y62, is approximately 25 Å (pdbID used 7TT3). The distance of the maximum amino acid length of the internal signal of OmpC, from F280 to Y288, is approximately 22 Å (pdbID used 2J1N). This would allow for the signal to fit within the confines of the TRP motif of BamD.

Author response image 1.



1. The authors highlight that the sequence motifs are common among the inhibiting peptides, but do not test if this is a necessary motif to mediate the interactions. It would have been good to see if a library of non-OMP related peptides that match this motif could also inhibit or not.

With respect, this additional work would not address any biological question relevant to the function of BamD. To randomize sequences and then classify those that do or don't fit the motif would help in refining the parameters of the β -signal motif, but that was not our intent.

We have identified the peptides from within the total sequence of an OMP, shown which peptides inhibit in an assembly assay, and then observed that the inhibitory peptides conform to a previously published (β -signal) motif.

1. In the studies that disrupt the motifs by mutagenesis, an effect was observed and attributed to disruption of the interaction of the 'internal signal'. However, the literature is filled with point mutations in OMPs that disrupt biogenesis, particular those within the membrane region. F280, Y286, V359, and Y365 are all residues that are in the membrane region that point into the membrane. Therefore, more work is needed to confirm that these mutations are in parts of a recognition motif rather than on the residues that are disrupting stability/assembly into the membrane.

As the reviewer pointed out, the side chains of the amino acids constituting the signal elements we determined were all facing the lipid side, of which Y286 and Y365 were important for folding as well as to be recognized. However, F280A and V359A had no effect on folding, but only on assembly through the BAM complex. The fact that position 0 functions as a signal has been demonstrated by peptidomimetics (Fig. 1) and point mutant analysis (Fig. 2). We appreciate this clarification and have modified the text to acknowledge that the all of the signal element faces the lipid side, which contributes to their stability in the membrane finally, and before that the BAM complex actively recognizes them and determines their orientation:

(Lines 519-526) After OMP assembly, all elements of the internal signal are positioned such that they face into the lipid-phase of the membrane. This observation may be a coincidence, or may be utilized by the BAM complex to register and orientate the lipid facing amino acids

in the assembling OMP away from the formative lumen of the OMP. Amino acids at position 6, such as Y286 in OmpC, are not only component of the internal signal for binding by the BAM complex, but also act in structural capacity to register the aromatic girdle for optimal stability of the OMP in the membrane.

1. The title of Figure 3 indicates that disrupting the internal signal motif disrupts OMP assembly, however, the point mutations did not seem to have any effect. Only when both 280 and 286 were mutated was an effect observed. And even then, the trimer appeared to form just fine, albeit at reduced levels, indicating assembly is just fine, rather the rate of biogenesis is being affected.

We appreciate this point and have revised the title of Figure 3 to be:

(Lines 1070-1071) "Modifications in the putative internal signal slow the rate of OMP assembly in vivo."

1. In Figure 4, the authors attempt to quantify their blots. However, this seems to be a difficult task given the lack of quality of the blots and the spread of the intended signals, particularly of the 'int' bands. However, the more disturbing trend is the obvious reduction in signal from the post-urea treatment, even for the WT samples. The authors are using urea washes to indicate removal of only stalled substrates. However a reduction of signal is also observed for the WT. The authors should quantify this blot as well, but it is clear visually that both WT and the mutant have obvious reductions in the observable signals. Further, this data seems to conflict with Fig 3D where no noticeable difference in OmpC assembly was observed between WT and Y286A, why is this the case?

We have addressed this point by adding a statistical analysis on Fig. 4A. As the reviewer points out, BN-PAGE band quantification is a difficult task given the broad spread of the bands on these gels. Statistical analysis showed that the increase in intermediates (int) was statistically significant for Y286A at all times until 80 min, when the intermediate form signals decrease.

(Lines 1093-1096) "Statistical significance was indicated by the following: N.S. (not significant), $p < 0.05$; *, $p < 0.005$; **. Exact p values of intermediate formed by Wt vs Y286A at each timepoint were as follows; 20 minutes: $p = 0.03077$, 40 minutes: $p = 0.02402$, 60 minutes: $p = 0.00181$, 80 minutes: $p = 0.0545$."

Further regarding the Int. band, we correct the statement as follows.

(Lines 253-254) "Consistent with this, the assembly intermediate which was prominently observed at the OmpC(Y286A) can be extracted from the membranes with urea;"

OMP assembly in vivo has additional periplasmic chaperones and factors present in order to support the assembly process. Therefore, it is likely that some proteins were assembled properly in vivo compared to their in vitro counterparts. Such a decrease has been observed not only in E. coli but also in mitochondrial OMP import (Yamano et al., 2010).

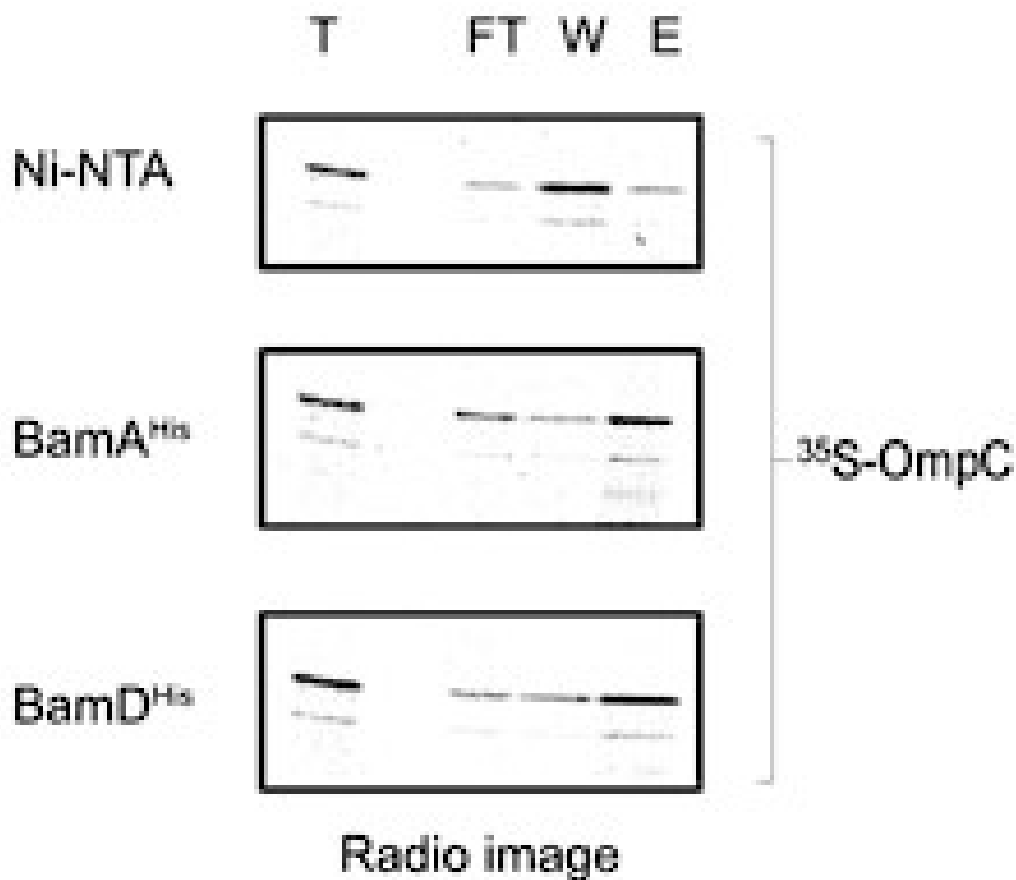
1. The pull-down assays with BamA and BamD should include a no protein control at the least to confirm there is no non-specific binding to the resin. Also, no detergent was mentioned as part of the pull downs that contained BamA or OmpC, nor was it detailed if OmpC was urea solubilized.

We have performed pull down experiments with a no-protein (Ni-NTA only) control as noted (Author response image 1). The results showed that the amount of OmpC carrying through on

beads only was significantly lower than the amount of OmpC bound in the presence of BamD or BamA. The added OmpC was not treated with urea, but was synthesized by in vitro translation; the in vitro translated OmpC is the standard substrate in the EMM assembly assay (Supp Fig. S1) where it is recognized by the BAM complex. Thus, we used it for pull-down as well and, to make this clearer, we have revised as follows:

Author response image 2.

Pull down assay of radio-labelled OmpC with indicated protein or Ni-NTA alone (Ni-NTA) . T; total, FT; Flow throw, W; wash, E; Elute.



(Lines 252-265) "Three subunits of the BAM complex have been previously shown to interact with the substrates: BamA, BamB, and BamD (Hagan et al., 2013; Harrison, 1996; Ieva et al., 2011). In vitro pull-down assay showed that while BamA and BamD can independently bind to the in vitro translated OmpC polypeptide (Fig .S9A), BamB did not (Fig. S9B)."

1.

- The neutron reflectometry experiments are not convincing primarily due to the lack controls to confirm a consistent uniform bilayer is being formed and even if so, uniform orientations of the BamA molecules across the surface.
- Further, no controls were performed with BamD alone, or with OmpC alone, and it is hard to understand how the method can discriminate between an actual BamA/BamD

complex versus BamA and BamD individually being located at the membrane surface without forming an actual complex.

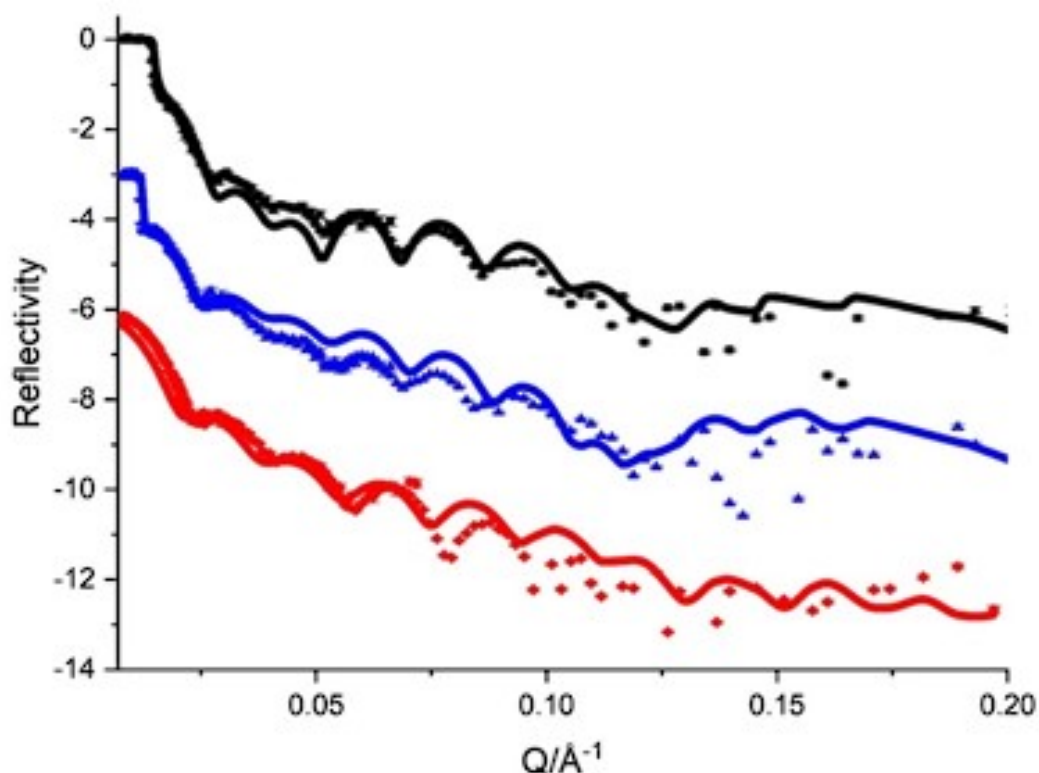
- Previous studies have reported difficulty in preparing a complex with BamA and BamD from purified components.
- Additionally, little signal differences were observed for the addition of OmpC. However, an elongated unfolded polypeptide that is nearly 400 residues long would be expected to produce a large distinct signal given that only the C-terminal portion is supposedly anchored to BAM, while the rest would be extended out above the surface.
- The depiction in Figure 5D is quite misleading when viewing the full structures on the same scales with one another.

We have addressed these five points individually as follows.

i. The uniform orientation of BamA on the surface is guaranteed by the fixation through a His-tag engineered into extracellular loop 6 of BamA and has been validated in previous studies as cited in the text. Moreover, to explain this, we reconstructed another theoretical model for BamA not oriented well in the system as below. However, we found that the solid lines (after fitting) didn't align well with the experimental data. We therefore assumed that BamA has oriented well in the membrane bilayer.

Author response image 3.

Experimental (symbols) and fitted (curves) NR profiles of BamA not oriented well in the POPC bilayer in D2O (black), GMW (blue) and H2O (red) buffer.

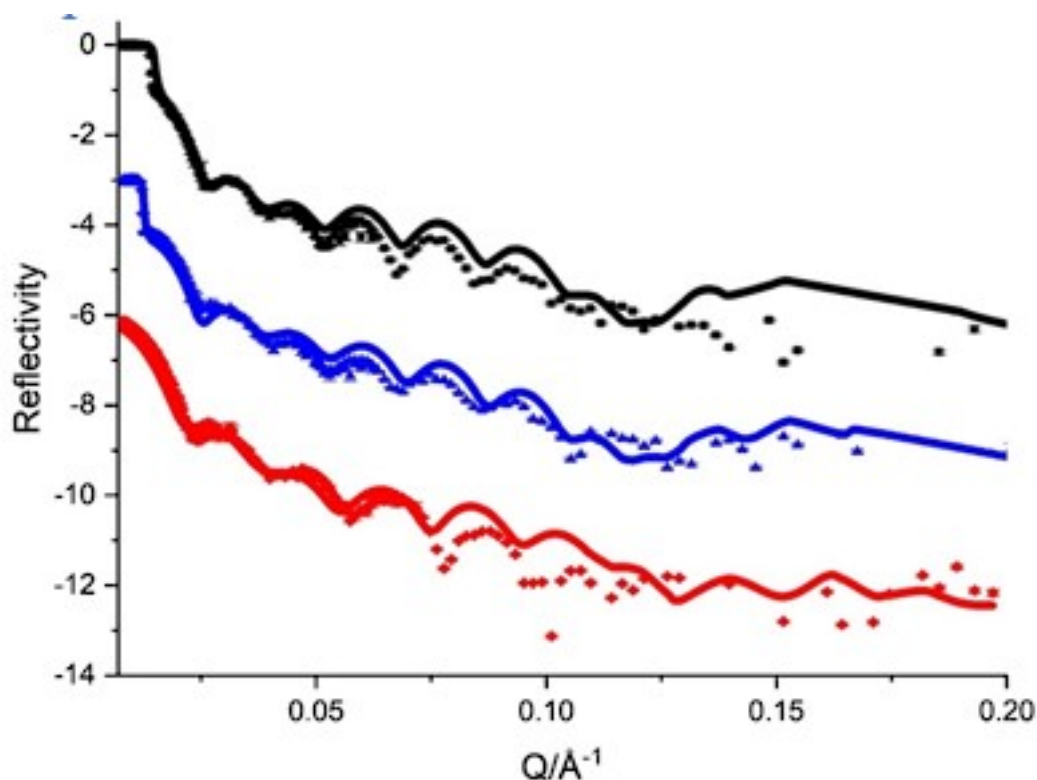


ii. There would be no means by which to do a control with OmpC alone or BamD alone as neither protein binds to the lipid layer chip. OmpC is diluted from urea and then the unbound OmpC is washed from the chip before NR measurements. BamD does not have an

acyl group to anchor it to the lipid layer, without BamA to anchor to, it too is washed from the chip before NR measurements. We have reconstructed another theoretical model for both of BamA + BamD embedding in the membrane bilayer, and the fits were shown below. Apparently, the fits didn't align well with the experimental data, which discriminate the BamA/BamD individually being located at the membrane surface without forming an actual complex.

Author response image 4.

Experimental (symbols) and fitted (curves) NR profiles of BamA+D embedding together in the POPC bilayer in D2O (black), GMW (blue) and H2O (red) buffer.



iii. The previous studies that reported difficulty in preparing a complex with BamA and BamD from purified components were assays done in aqueous solution including detergent solubilized BamA, or with BamA POTRA domains only. Our assay is superior in that it reports the binding of BamD to a purified BamA that has been reconstituted in a lipid bilayer.

iv. The relatively small signal differences observed for the addition of OmpC are expected, since OmpC is an elongated, unfolded polypeptide of nearly 400 residues long which, in the context of this assay, can occupy a huge variation in the positions at which it will sit with only the C-terminal portion anchored to BAM, and the rest moving randomly about and extended from the surface.

v. We appreciate the point raised and have now added a note in the Figure legend that these are depictions of the results and not a scale drawing of the structures.

1. In the crosslinking studies, the authors show 17 crosslinking sites (43% of all tested) on BamD crosslinked with OmpC. Given that the authors are presenting specific interactions between the two proteins, this is worrisome as the crosslinks were found across the entire surface of BamD. How do the authors explain this? Are all these specific or non-specific?

The crosslinking experiment using purified BamD was an effective assay for comprehensive analysis of the interaction sites between BamD and the substrate. However, as the reviewer pointed out, cross-linking was observed even at the sites that, in the context of the BAM complex, interact with BamC as a protein-protein interaction and would not be available for substrate protein-protein interactions. To complement this, analysis and to address this issue, we also performed the experiment in Fig. 6C.

In Fig. 6C, the interaction of BamD with the substrate is examined *in vivo*, and the results demonstrate that if BPA is introduced into the site, we designated as the substrate recognition site, it is cross-linked to the substrate. On the other hand, position 114 was found to crosslink with the substrate *in vitro* crosslinking, but not *in vivo*. It should be noted that position 114 has also been confirmed to form cross-link products with BamC, we believe that BamD-substrate interactions in the native state have been investigated. To explain the above, we have added the following description to the Results section.

(Lines 319-321) "Structurally, these amino acids locate both the lumen side of funnel-like structure (e.g. 49 or 62) and outside of funnel-like structure such as BamC binding site (e.g. 114) (fig. S12C). (Lines 350-357) Positions 49, 53, 65, and 196 of BamD face the interior of the funnel-like structure of the periplasmic domain of the BAM complex, while position 114 is located outside of the funnel-like structure (Bakelar et al., 2016; Gu et al., 2016; Iadanza et al., 2016). We note that while position 114 was cross-linked with OmpC *in vitro* using purified BamD, that this was not seen with *in vivo* cross-linking. Instead, in the context of the BAM complex, position 114 of BamD binds to the BamC subunit and would not be available for substrate binding *in vivo* (Bakelar et al., 2016; Gu et al., 2016; Iadanza et al., 2016)."

1. The study in Figure 6 focuses on defined regions within the OmpC sequence, but a more broad range is necessary to demonstrate specificity to these regions vs binding to other regions of the sequence as well. If the authors wish to demonstrate a specific interaction to this motif, they need to show no binding to other regions.

The region of affinity for the BAM complex was determined by peptidomimetic analysis, and the signal region was further identified by mutational analysis of OmpC. Subsequently, the subunit that recognizes the signal region was identified as BamD. In other words, in the process leading up to Fig. 6, we were able to analyze in detail that other regions were not the target of the study. We have revised the text to make clear that we focus on the signal region including the internal signal, and have not also analyzed other parts of the signal region:

(Lines 329-332) "As our peptidomimetic screen identified conserved features in the internal signal, and cross-linking highlighted the N-terminal and C-terminal TPR motifs of BamD as regions of interaction with OmpC, we focused on amino acids specifically within the β -signals of OmpC and regions of BamD which interact with β -signal."

1. The levels of the crosslinks are barely detectable via western blot analysis. If the interactions between the two surfaces are required, why are the levels for most of the blots so low?

These are western blots of cross-linked products – the efficiency of cross-linking is far less than 100% of the interacting protein species present in a binding assay and this explains why the levels for the blots are ‘so low’. We have added a sentence to the revised manuscript to make this clear for readers who are not molecular biologists:

(Lines 345-348) "These western blots reveal cross-linked products representing the interacting protein species. Photo cross-linking of unnatural amino acid is not a 100% efficient process, so the level of cross-linked products is only a small proportion of the molecules interacting in the assays."

1.

- *Figure 7 indicates that two regions of BamD promote OMP orientation and assembly, however, none of the experiments appears to measure OMP orientation?*
- *Also, one common observation from panel F was that not only was the trimer reduced, but also the monomer. But even then, still a percentage of the trimer is formed, not a complete loss.*

(i) We appreciate this point and have revised the title of Figure 7 to be:

(Lines 1137-1138) "Key residues in two structurally distinct regions of BamD promote β -strand formation and OMP assembly."

(ii) In our description of Fig. 7F (Lines 356-360) we do not distinguish between the amount of monomer and trimer forms, since both are reflective of the overall assembly rate i.e. assembly efficiency. Rather, we state that:

"The EMM assembly assay showed that the internal signal binding site was as important as the β -signal binding site to the overall assembly rates observed for OmpC (Fig. 7F), OmpF (fig. S15D), and LamB (fig. S15E). These results suggest that recognition of both the C-terminal β -signal and the internal signal by BamD is important for efficient protein assembly."

1.

- *The experiment in Fig 7B would be more conclusive if it was repeated with both the Y62A and R197A mutants and a double mutant. These controls would also help resolve any effect from crowding that may also promote the crosslinks.*
- *Further, the mutation of R197 is an odd choice given that this residue has been studied previously and was found to mediate a salt bridge with BamA. How was this resolved by the authors in choosing this site since it was not one of the original crosslinking sites?*

As stated in the text, the purpose of the experiment in Figure 7B is to measure the impact of pre-forming a β -strand in the substrate (OmpC) before providing it to the receptor (BamD). We thank the reviewer for the comment on the R197 position of BamD. The C-terminal domain of BamD has been suggested to mediate the BamA-BamD interface, specifically BamD R197 amino acid creates a salt-bridge with BamA E373 (Ricci et al., 2012). It had been postulated that the formation of this salt-bridge is not strictly structural, with R197 highlighted as a key amino acid in BamD activity and this salt-bridge acts as a “check-point” in BAM complex activity (Ricci et al., 2012, Storek et al., 2023). Our results agree with this, showing that the C-terminus of BamD acts in substrate recognition and alignment of the β -signal (Fig. 6, Fig S12). We show that amino acids in the vicinity of R197 (N196, G200, D204) cross-linked well to substrate and mutations to the β -signal prevent this interaction (Fig S12B, D). For mutational analysis of BamD, we looked then at the conservation of the C-terminus of

BamD and determined R197 was the most highly conserved amino acid (Fig 6C). In order to account for this, we have adjusted the manuscript:

(Lines 376-377) "R197 has previously been isolated as a suppressor mutation of a BamA temperature sensitive strain (Ricci et al., 2012)."

(Lines 495-496) "This adds an additional role of the C-terminus of BamD beyond a complex stability role (Ricci et al., 2012; Storek et al., 2023)."

1. *As demonstrated by the authors in Fig 8, the mutations in BamD lead to reduction in OMP levels for more than just OmpC and issues with the membrane are clearly observable with Y62A, although not with R197A in the presence of VCN. The authors should also test with rifampicin which is smaller and would monitor even more subtle issues with the membrane. Oddly, no growth was observed for the Vec control in the lower concentration of VCN, but was near WT levels for 3 times VCN, how is this explained?*

While it would be interesting to correlate the extent of differences to the molecular size of different antibiotics such as rifampicin, such correlations are not the intended aim of our study. Vancomycin (VCN) is a standard measure of outer membrane integrity in our field, hence its use in our tests for membrane integrity.

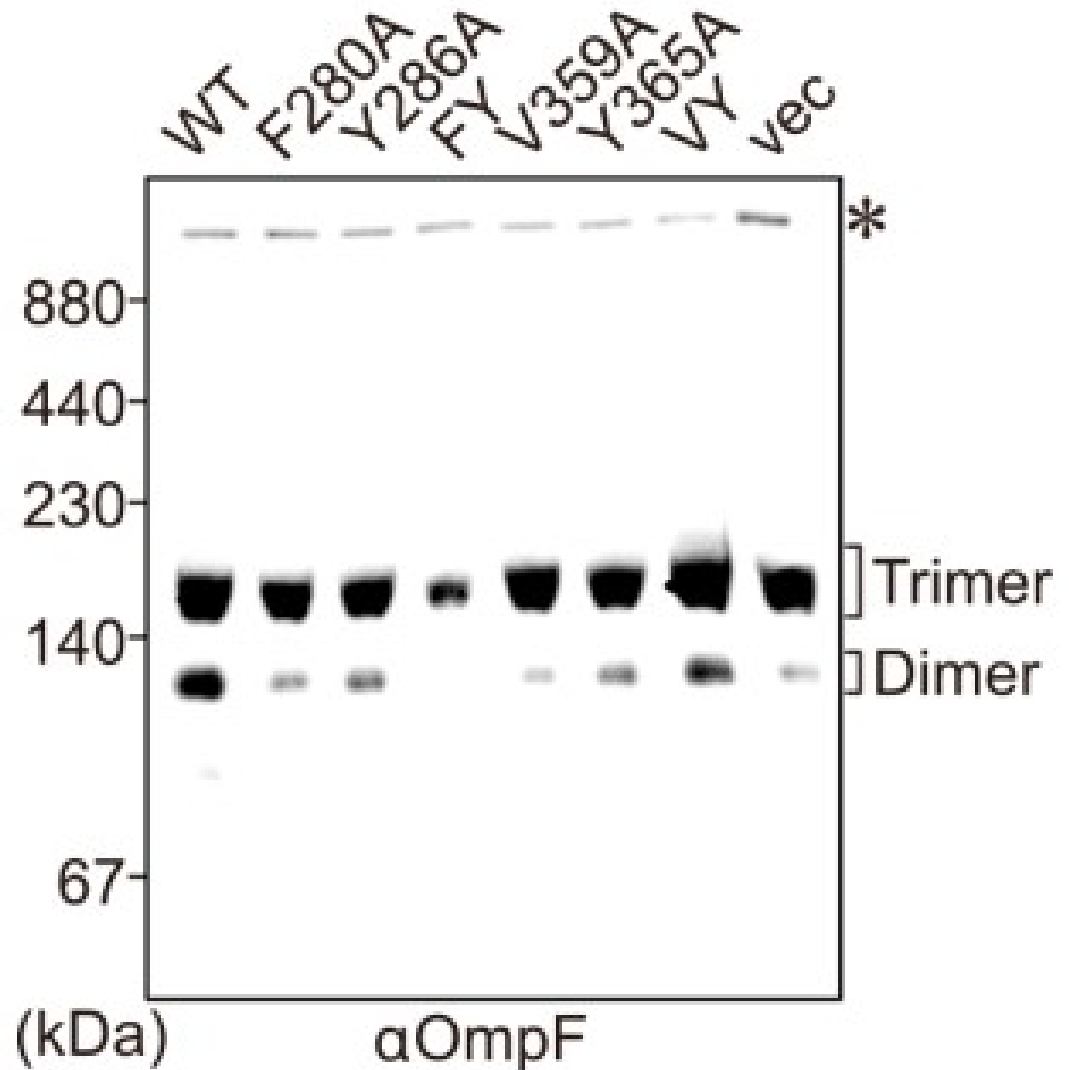
We apologize to the reviewer as Figure 8 D-G may have been misleading. Figure 8D,E are using bamD shut-down cells expressing plasmid-borne BamD mutants. Whereas Figure 8F, G are the same strain as used in Figure 3. We have adjusted the figure as well as the figure legend: (Lines 1165-1169) D, E E coli bamD depletion cells expressing mutations at residues, Y62A and R197A, in the β -signal recognition regions of BamD were grown with of VCN. F, G, E coli cells expressing mutations to OmpC internal signal, as shown in Fig 3, grown in the presence of VCN. Mutations to two key residues of the internal signal were sensitive to the presence of VCN.

1. *While Fig 8I indeed shows diminished levels for FY as stated, little difference was observed for the trimer for the other mutants compared to WT, although differences were observed for the dimer. Interestingly, the VY mutant has nearly WT levels of dimer. What do the authors postulate is going on here with the dimer to trimer transition? How do the levels of monomer compare, which is not shown?*

The BN-PAGE gel system cannot resolve protein species that migrate below ~50kDa and the monomer species of the OMPs is below this size. We can't comment on effects on the monomer because it is not visualized. The non-cropped gel image is shown here. Recently, Hussain et al., has shown that in vitro proteo-liposome system OmpC assembly progresses from a "short-lived dimeric" form before the final process of trimerization (Hussain et al., 2021). However, their findings suggest that LPS plays the final role in stimulation of dimer-to-trimer, a step well past the recognition step of the β -signals. Mutations to the internal signal of OmpC results in the formation of an intermediate, the substrate stalled on the BAM complex. This stalling, presumably, causes a hinderance to the BAM complex resulting in reduced timer and loss of dimer OmpF signal in the EMM of cells expressing OmpC double mutant strain, FY. cannot resolve protein species that migrate below ~50kDa and the monomer species of the OMPs is below this size. We can't comment on effects on the monomer because it is not visualized. The non-cropped gel image is shown here. We have noted this in the revised text:

Author response image 5.

Non-cropped gel of Fig. 8I. the asterisk indicates a band observed in the sample loading wells at the top of the gel.



(Lines 417-418) "The dimeric form of endogenous OmpF was prominently observed in both the OmpC(WT) as well as the OmpC(VY) double mutant cells."

1. In the discussion, the authors indicate they have '...defined an internal signal for OMP assembly', however, their study is limited and only investigates a specific region of OmpC. More is needed to definitively say this for even OmpC, and even more so to indicate this is a general feature for all OMPs.

We acknowledge the reviewer's comment on this point and have expanded the statement to make sure that the conclusion is justified with the specific evidence that is shown in the paper and the supplementary data. We now state:

(Lines 444-447) "This internal signal corresponds to the -5 strand in OmpC and is recognized by BamD. Sequence analysis shows that similar sequence signatures are present in other OMPs (Figs. S5, S6 and S7). These sequences were investigated in two further OMPs: OmpF and LamB (Fig. 2C and D)."

Note, we did not state that this is a general feature for all OMPs. That would not be a reasonable proposition.

1.

- *In the proposed model in Fig 9, it is hard to conceive how 5 strands will form along BamD given the limited surface area and tight space beneath BAM.*
- *More concerning is that the two proposal interaction sites on BamD, Y62 and R197, are on opposite sides of the BamD structure, not along the same interface, which makes this model even more unlikely.*
- *As evidence against this model, in Figure 9E, the two indicates sites of BamD are not even in close proximity of the modeled substrate strands.*

We can address the reviewer's three concerns here:

i. The first point is that the region (formed by BamD engaged with POTRA domains 1-2 and 5 of BamA) is not sufficient to accommodate five β -strands. Structural analysis reveals that the interaction between the N-terminal side of BamD and POTRA1-2 is substantially changed the conformation by substrate binding, and that this surface is greatly extended. This surface does have enough space to accommodate five beta-strands, as now documented in Fig. 9D, 9E using the latest structures (7TT5 and 7TT2) as illustrations of this. The text now reads:

(Lines 506-515) "Spatially, this indicates the BamD can serve to organize two distinct parts of the nascent OMP substrate at the periplasmic face of the BAM complex, either prior to or in concert with, engagement to the lateral gate of BamA. Assessing this structurally showed the N-terminal region of BamD (interacting with the POTRA1-2 region of BamA) and the C-terminal region of BamD (interacting with POTRA5 proximal to the lateral gate of BamA) (Bakelar et al., 2016; Gu et al., 2016; Tomasek et al., 2020) has the N-terminal region of BamD changing conformation depending on the folding states of the last four β -strands of the substrate OMP, EspP (Doyle et al., 2022). The overall effect of this being a change in the dimensions of this cavity change, a change which is dependent on the folded state of the substrate engaged in it (Fig 9 B-E)."

ii. The second point raised regards the orientation of the substrate recognition residues of BamD. Both Y62A and R197 were located on the lumen side of the funnel in the EspP-BAM transport intermediate structure (PDBID; 7TTC); Y62A is relatively located on the edge of BamD, but given that POTRA1-2 undergoes a conformational change and opens this region, as described above, both are located in locations where they could bind to substrates. This was explained in the following text in the results section of revised manuscript.

(Lines 377-379) "Each residue was located on the lumen side of the funnel-like structure in the EspP-BAM assembly intermediate structure (PDBID; 7TTC) (Doyle et al., 2022)."

****Reviewer #2 (Public Review):"**

Previously, using bioinformatics study, authors have identified potential sequence motifs that are common to a large subset of beta-barrel outer membrane proteins in gram negative bacteria. Interestingly, in that study, some of those motifs are located in the

internal strands of barrels (not near the termini), in addition to the well-known "beta-signal" motif in the C-terminal region.

Here, the authors carried out rigorous biochemical, biophysical, and genetic studies to prove that the newly identified internal motifs are critical to the assembly of outer membrane proteins and the interaction with the BAM complex. The author's approaches are rigorous and comprehensive, whose results reasonably well support the conclusions. While overall enthusiastic, I have some scientific concerns with the rationale of the neutron refractory study, and the distinction between "the intrinsic impairment of the barrel" vs "the impairment of interaction with BAM" that the internal signal may play a role in. I hope that the authors will be able to address this.

Strengths:

- 1. It is impressive that the authors took multi-faceted approaches using the assays on reconstituted, cell-based, and population-level (growth) systems.*
- 1. Assessing the role of the internal motifs in the assembly of model OMPs in the absence and presence of BAM machinery was a nice approach for a precise definition of the role.*

Weaknesses:

- 1. The result section employing the neutron refractory (NR) needs to be clarified and strengthened in the main text (from line 226). In the current form, the NR result seems not so convincing.*

What is the rationale of the approach using NR?

We have now modified the text to make clear that:

(Lines 276-280) "The rationale to these experiments is that NR provides: (i) information on the distance of specified subunits of a protein complex away from the atomically flat gold surface to which the complex is attached, and (ii) allows the addition of samples between measurements, so that multi-step changes can be made to, for example, detect changes in domain conformation in response to the addition of a substrate."

What is the molecular event (readout) that the method detects?

We have now modified the text to make clear that:

(Lines 270-274) "While the biochemical assay demonstrated that the OmpC(Y286A) mutant forms a stalled intermediate with the BAM complex, in a state in which membrane insertion was not completed, biochemical assays such as this cannot elucidate where on BamA-BamD this OmpC(Y286A) substrate is stalled."

What are "R"-y axis and "Q"-x axis and their physical meanings (Fig. 5b)?

The neutron reflectivity, R , refers to the ratio of the incoming and exiting neutron beams and it is measured as a function of Momentum transfer Q , which is defined as $Q=4\pi \sin\theta/\lambda$, where θ is the angle of incident and λ is the neutron wavelength. $R(Q)$ is approximately given by $R(Q)=16\pi^2/Q^2 |\rho(Q)|^2$, where $R(Q)$ is the one-dimensional Fourier transform of $\rho(z)$, the scattering length density (SLD) distribution normal to the surface. SLD is the sum of the coherent neutron scattering lengths of all atoms in the sample layer divided by the volume of the layer. Therefore, the intensity of the reflected beams is highly dependent on the thickness,

densities and interface roughness of the samples. This was explained in the following text in the method section of revised manuscript.

(Lines 669-678) "Neutron reflectivity, denoted as R , is the ratio of the incoming to the exiting neutron beams. It's calculated based on the Momentum transfer Q , which is defined by the formula $Q=4\pi \sin\theta/\lambda$, where θ represents the angle of incidence and λ stands for the neutron wavelength. The approximate value of $R(Q)$ can be expressed as $R(Q)=16\pi^2/Q^2 |\rho(Q)|^2$, where $R(Q)$ is the one-dimensional Fourier transform of $\rho(z)$, which is the scattering length density (SLD) distribution perpendicular to the surface. SLD is calculated by dividing the sum of the coherent neutron scattering lengths of all atoms in a sample layer by the volume of that layer. Consequently, factors such as thickness, volume fraction, and interface roughness of the samples significantly influence the intensity of the reflected beams."

| *How are the "layers" defined from the plot (Fig. 5b)?*

The "layers" in the plot (Fig. 5b) represent different regions of the sample being studied. In this study, we used a seven-layer model to fit the experimental data (chromium - gold - NTA - HIS8 - β -barrel - P3-5 - P1-2. This was explained in the following text in the figure legend of revised manuscript. (Lines 1115-1116) The experimental data was fitted using a seven-layer model: chromium - gold - NTA - His8 - β -barrel - P3-5 - P1-2.

| *What are the meanings of "thickness" and "roughness" (Fig. 5c)?*

We used neutron reflectometry to determine the relative positions of BAM subunits in a membrane environment. The binding of certain subunits induced conformational changes in other parts of the complex. When a substrate membrane protein is added, the periplasmic POTRA domain of BamA extends further away from the membrane surface. This could result in an increase in thickness as observed in neutron reflectometry measurements.

As for roughness, it is related to the interface properties of the sample. In neutron reflectometry, the intensity of the reflected beams is highly dependent on the thickness, densities, and interface roughness of the samples. An increase in roughness could suggest changes in these properties, possibly due to protein-membrane interactions or structural changes within the membrane.

(Lines 1116-1120) "Table summarizes of the thickness, roughness and volume fraction data of each layer from the NR analysis. The thickness refers to the depth of layered structures being studied as measured in Å. The roughness refers to the irregularities in the surface of the layered structures being studied as measured in Å."

| *What does "SLD" stand for?*

We apologize for not explaining abbreviation when the SLD first came out. We explained it in revised manuscript. (Line 298)

| *1. In the result section, "The internal signal is necessary for insertion step of assembly into OM" This section presents an important result that the internal beta-signal is critical to the intrinsic propensity of barrel formation, distinct from the recognition by BAM complex. However, this point is not elaborated in this section. For example, what is the role of these critical residues in the barrel structure formation? That is, are they involved in any special tertiary contacts in the structure or in membrane anchoring of the nascent polypeptide chains?*

We appreciate the reviewer's comment on this point. Both position 0 and position 6 appear to be important amino acids for recognition by the BAM complex, since mutations introduced at

these positions in peptide 18 prevent competitive inhibition activity.

In terms of the tertiary structure of OmpC, position 6 is an amino acid that contributes to the aromatic girdle, and since Y286A and Y365A affected OMP folding as measured in folding experiments, it is perhaps their position in the aromatic girdle that contributes to the efficiency of β -barrel folding in addition to its function as a recognition signal. We have added a sentence in the revised manuscript:

(Lines 233-236) "Position 6 is an amino acid that contributes to the aromatic girdle. Since Y286A and Y365A affected OMP folding as measured in folding experiments, their positioning into the aromatic girdle may contribute to the efficiency of β -barrel folding, in addition to contributing to the internal signal."

The mutations made at position 0 had no effect on folding, so this residue may function solely in the signal. Given the register of each β -strand in the final barrel, the position 0 residues have side-chains that face out into the lipid environment. From examination of the OmpC crystal structure, the residue at position 0 makes no special tertiary contacts with other, neighbouring residues.

Reviewer #1 (Recommendations For The Authors):

Minor critiques (in no particular order):

1. Peptide 18 was identified based on its strong inhibition for EspP assembly but another peptide, peptide 23, also shows inhibition and has no particular consensus.

We would correct this point. Peptide 23 has a strong consensus to the canonical β -signal. We had explained the sequence consensus of β -signal in the Results section of the text. In the third paragraph, we have added a sentence indicating the relationship between peptide 18 and peptide 23.

(Lines 152-168) "Six peptides (4, 10, 17, 18, 21, and 23) were found to inhibit EspP assembly (Fig. 1A). Of these, peptide 23 corresponds to the canonical β -signal of OMPs: it is the final β -strand of OmpC and it contains the consensus motif of the β -signal (ζ xGxx[Ω / Φ]x[Ω / Φ]). The inhibition seen with peptide 23 indicated that our peptidomimetics screening system using EspP can detect signals recognized by the BAM complex. In addition to inhibiting EspP assembly, five of the most potent peptides (4, 17, 18, 21, and 23) inhibited additional model OMPs; the porins OmpC and OmpF, the peptidoglycan-binding OmpA, and the maltoporin LamB (fig. S3). Comparing the sequences of these inhibitory peptides suggested the presence of a sub-motif from within the β -signal, namely [Ω / Φ]x[Ω / Φ] (Fig. 1B). The sequence codes refer to conserved residues such that: ζ , is any polar residue; G is a glycine residue; Ω is any aromatic residue; Φ is any hydrophobic residue and x is any residue (Hagan et al., 2015; Kutik et al., 2008). The non-inhibitory peptide 9 contained some elements of the β -signal but did not show inhibition of EspP assembly (Fig. 1A).

Peptide 18 also showed a strong sequence similarity to the consensus motif of the β -signal (Fig. 1B) and, like peptide 23, had a strong inhibitory action on EspP assembly (Fig. 1A). Variant peptides based on the peptide 18 sequence were constructed and tested in the EMM assembly assay (Fig. 1C)."

1. It is unclear why the authors immediately focused on BamD rather than BamB, given that both were mentioned to mediate interaction with substrate. Was BamB also tested?

We thank the reviewer for this comment. Following the reviewer's suggestion, we have now performed a pull-down experiment on BamB and added it to Fig. S9. We also modified the

text of the results as follows.

(Lines 262-265) "Three subunits of the BAM complex have been previously shown to interact with the substrates: BamA, BamB, and BamD (Hagan et al., 2013; Harrison, 1996; Ieva et al., 2011). In vitro pull-down assay showed that while BamA and BamD can independently bind to the in vitro translated OmpC polypeptide (Fig. S9A), BamB did not (Fig. S9B)."

1. *For the in vitro folding assays of the OmpC substrates, labeled and unlabeled, no mention of adding SurA or any other chaperone which is known to be important for mediating OMP biogenesis in vitro.*

We appreciate the reviewer's concerns on this point, however chaperones such as SurA are non-essential factors in the OMP assembly reaction mediated by the BAM complex: the surA gene is not essential and the assembly of OMPs can be measured in the absence of exogenously added SurA. It remains possible that addition of SurA to some of these assays could be useful in detailing aspects of chaperone function in the context of the BAM complex, but that was not the intent of this study.

1. *For the supplementary document, it would be much easier for the reader to have the legends groups with the figures.*

Following the reviewer's suggestion, we have placed the legends of Supplemental Figures together with each Figure.

1. *Some of the figures and their captions are not grouped properly and are separated which makes it hard to interpret the figures efficiently.*

We thank the reviewer for this comment, we have revised the manuscript and figures to properly group the figures and captions together on a single page.

1. *The authors begin their 'Discussion' with a question (line 454), however, they don't appear to answer or even attempt to address it; suggest removing rhetorical questions.*

As per the reviewers' suggestion, we removed this question.

1. *Line 464, 'unbiased' should be removed. This would imply that if not stated, experiments are 'negatively' biased.*

We removed this word and revised the sentence as follows:

(Lines 431-433) "In our experimental approach to assess for inhibitory peptides, specific segments of the major porin substrate OmpC were shown to interact with the BAM complex as peptidomimetic inhibitors."

1. *Lines 466-467; '...go well beyond expected outcomes.' What does this statement mean?*

Our peptidomimetics led to unexpected results in elucidating the additional essential signal elements. The manuscript was revised as follows:

(Lines 433-435) "Results for this experimental approach went beyond expected outcomes by identifying the essential elements of the signal $\Phi\text{xxxxxx}[\Omega/\Phi]\text{x}[\Omega/\Phi]$ in β -strands other than the C-terminal strand."

1. Line 478; '...rich information that must be oversimplified...'?

We appreciate the reviewer's pointed out. For more clarity, the manuscript was revised as follows:

(Lines 450-453) "The abundance of information which arises from modeling approaches and from the multitude of candidate OMPs, is generally oversimplified when written as a primary structure description typical of the β -signal for bacterial OMPs (i.e. $\zeta xGxx[\Omega/\Phi]x[\Omega/\Phi]$) (Kutik et al., 2008)."

1. There are typos in the supplementary figures.

We have revised and corrected the Supplemental Figure legends.

Reviewer #2 (Recommendations For The Authors):

1. In Supplementary Information, I recommend adding the figure legends directly to the corresponding figures. Currently, it is very inconvenient to go back and forth between legends and figures.

Following the reviewer's suggestion, we have placed the legends of Supplemental Figures together with each Figure.

1. Line 94 (p.3): "later"

Lateral?

Yes. We have corrected this.

1. Line 113 (p.3): The result section, "Peptidomimetics derived from *E. coli* OmpC inhibit OMP assembly" Rationale of the peptide inhibition assay is not clear. How can the peptide sequence that effectively inhibit the assembly interpreted as the b-assembly signal? By competitive binding to BAM or by something else? What is the authors' hypothesis in doing this assay?

In revision, we have added following sentence to explain the aim and design of the peptidomimetics:

(Lines 140-145) "The addition of peptides with BAM complex affinity, such as the OMP β -signal, are capable of exerting an inhibitory effect by competing for binding of substrate OMPs to the BAM complex (Hagan et al., 2015). Thus, the addition of peptides derived from the entirety of OMPs to the EMM assembly assay, which can evaluate assembly efficiency with high accuracy, expects to identify novel regions that have affinity for the BAM complex."

1. Line 113- (p.3) and Fig. S1: The result section, "Peptidomimetics derived from *E. coli* OmpC inhibit OMP assembly"

Some explanation seems to be needed why b-barrel domain of EspP appears even without ProK?

We appreciate the reviewer's pointed out. We added following sentence to explain:

(Lines 128-137) "EspP, a model OMP substrate, belongs to autotransporter family of proteins. Autotransporters have two domains; (1) a β -barrel domain, assembled into the outer membrane via the BAM complex, and (2) a passenger domain, which traverses the outer membrane via the lumen of the β -barrel domain itself and is subsequently cleaved by the correctly assembled β -barrel domain (Celik et al., 2012). When EspP is correctly assembled into outer membrane, a visible decrease in the molecular mass of the protein is observed due to the self-proteolysis. Once the barrel domain is assembled into the membrane it becomes protease-resistant, with residual unassembled and passenger domains degraded (Leyton et al., 2014; Roman-Hernandez et al., 2014)."

1. Line 186 (p.6): "Y285"

Y285A?

We have corrected the error, it was Y285A.

1. Lines 245- (p. 7)/ Lines 330- (p. 10)

It needs to be clarified that the results described in these paragraphs were obtained from the assays with EMM.

We appreciate the reviewer's concerns on these points. For the first half, the following text was added at the beginning of the applicable paragraph to indicate that all of Fig. 4 is the result of the EMM assembly assay.

(Line 241) "We further analyzed the role of internal β -signal by the EMM assembly assay. At the second half, we used purified BamD but not EMM. We described clearly with following sentence."

(Lines 316-318) "We purified 40 different BPA variants of BamD, and then irradiated UV after incubating with 35S-labelled OmpC."